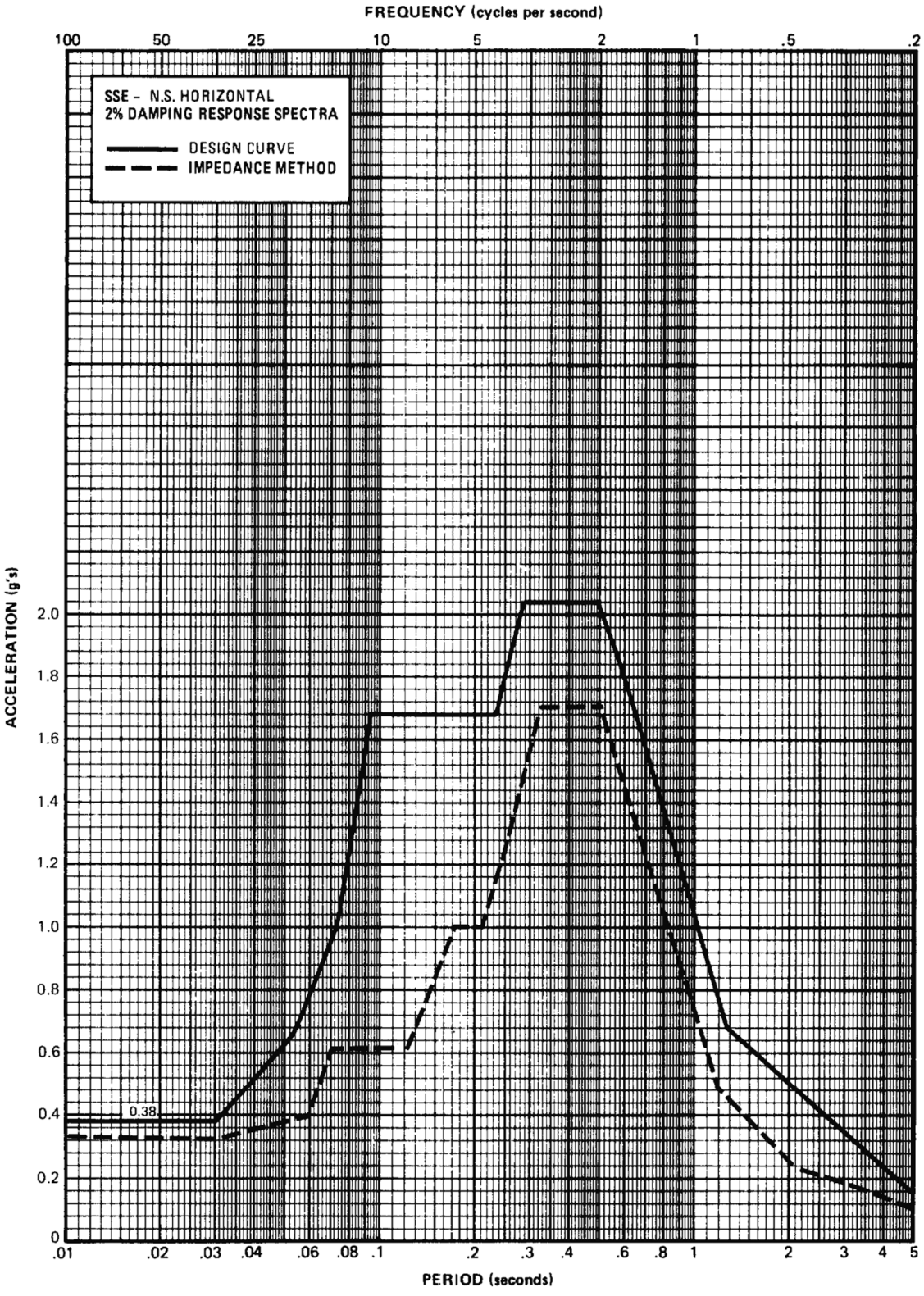


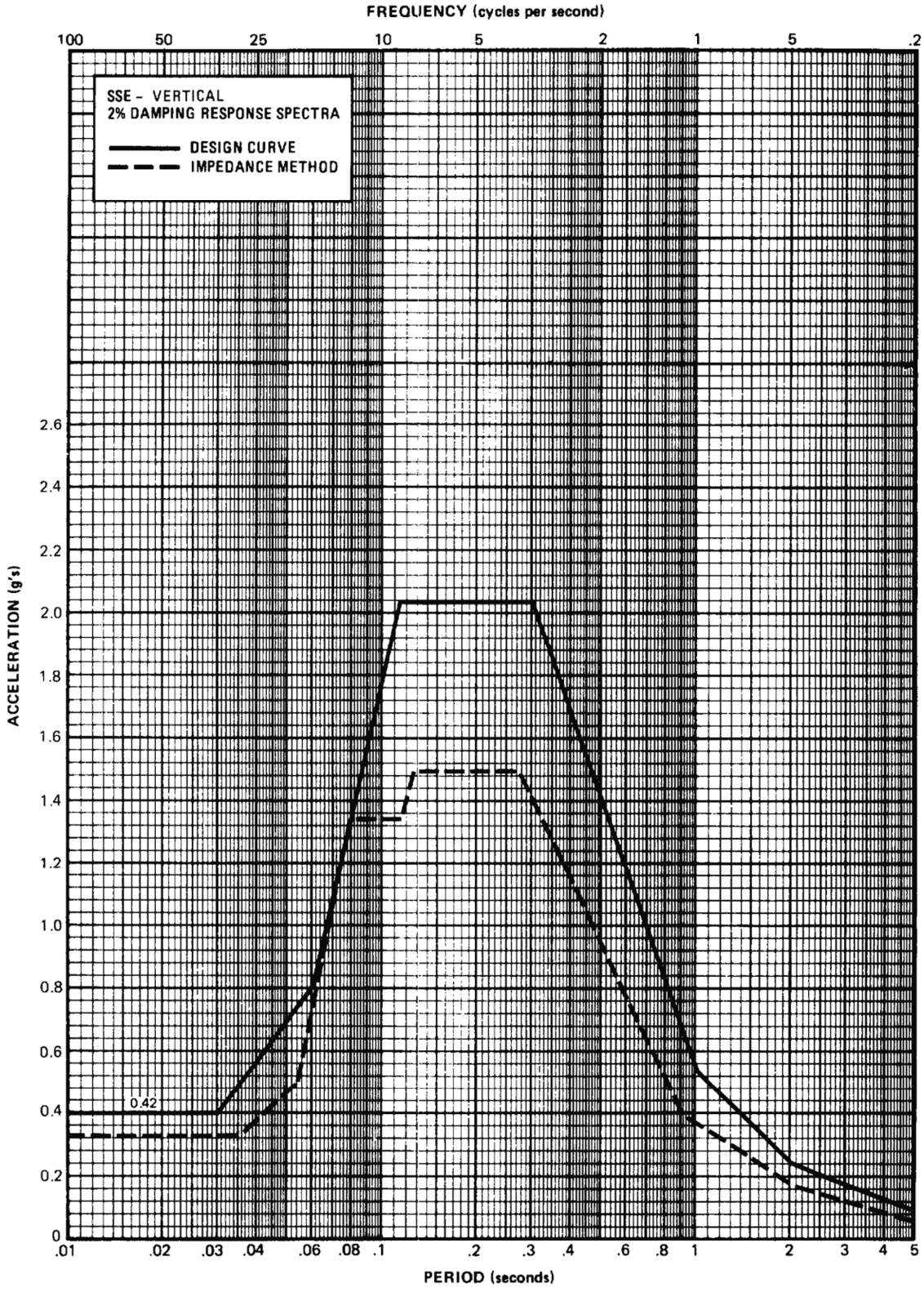

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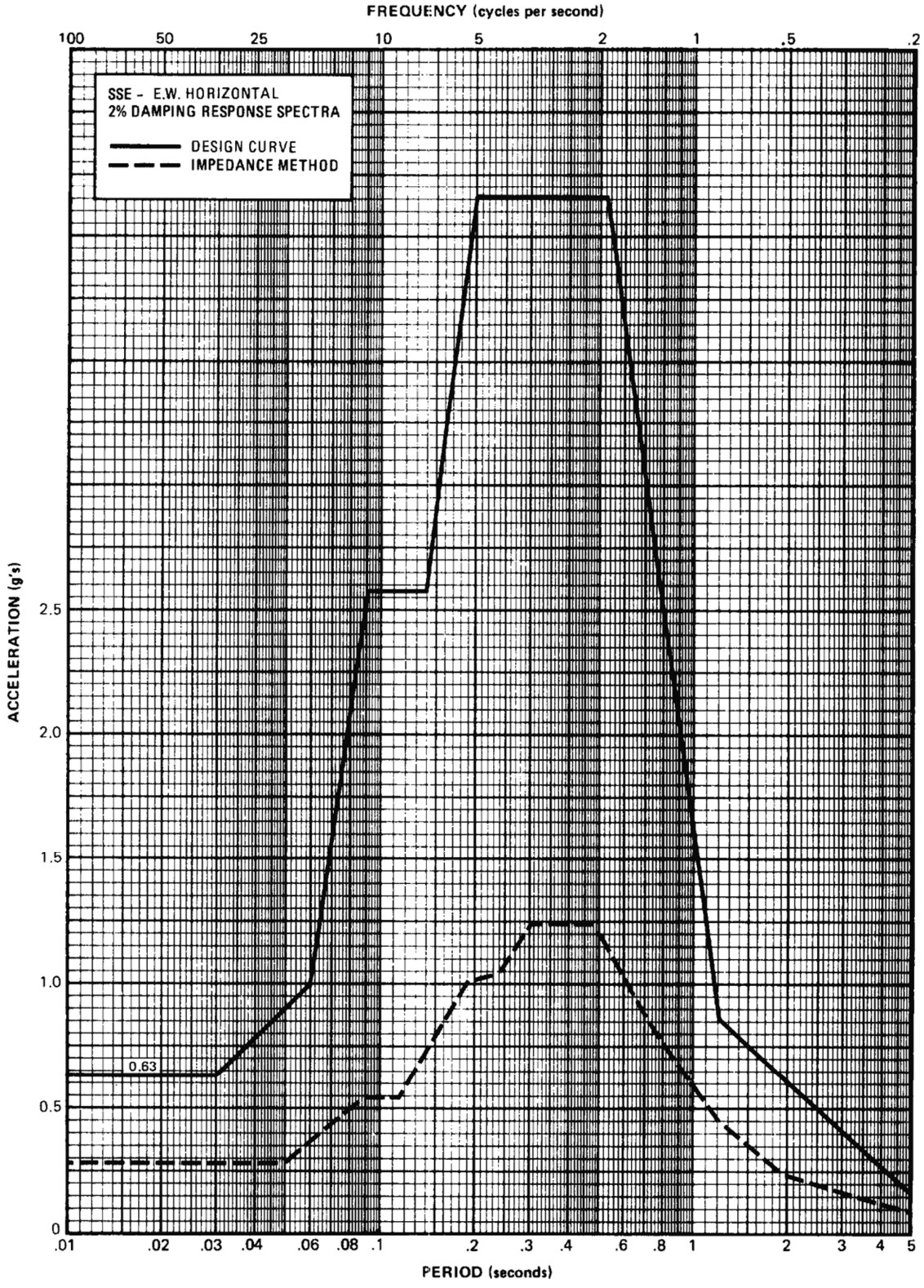
CONTROL BUILDING
EL. 220 FT.

FIGURE 3D–144



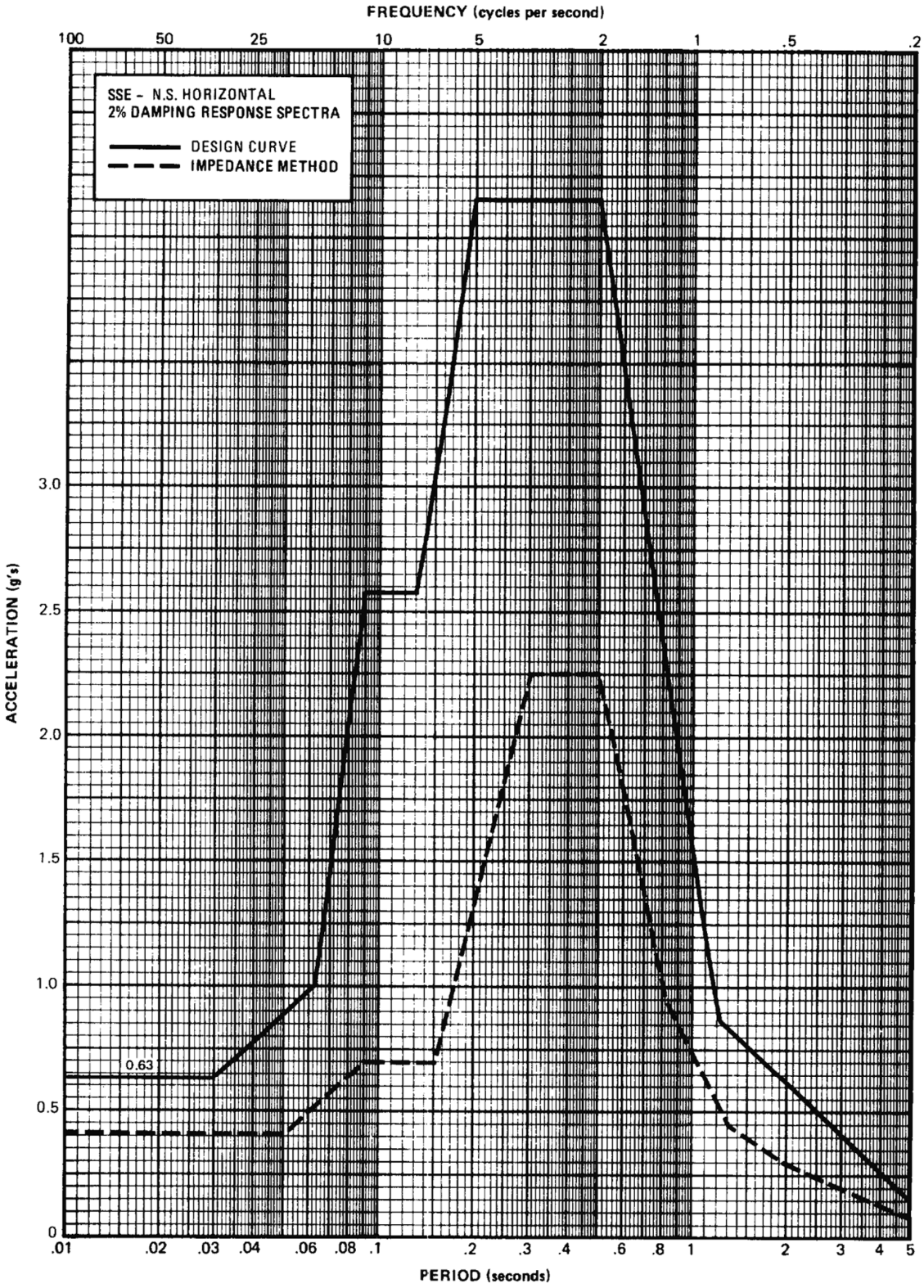

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| | FIGURE 3D–145 |



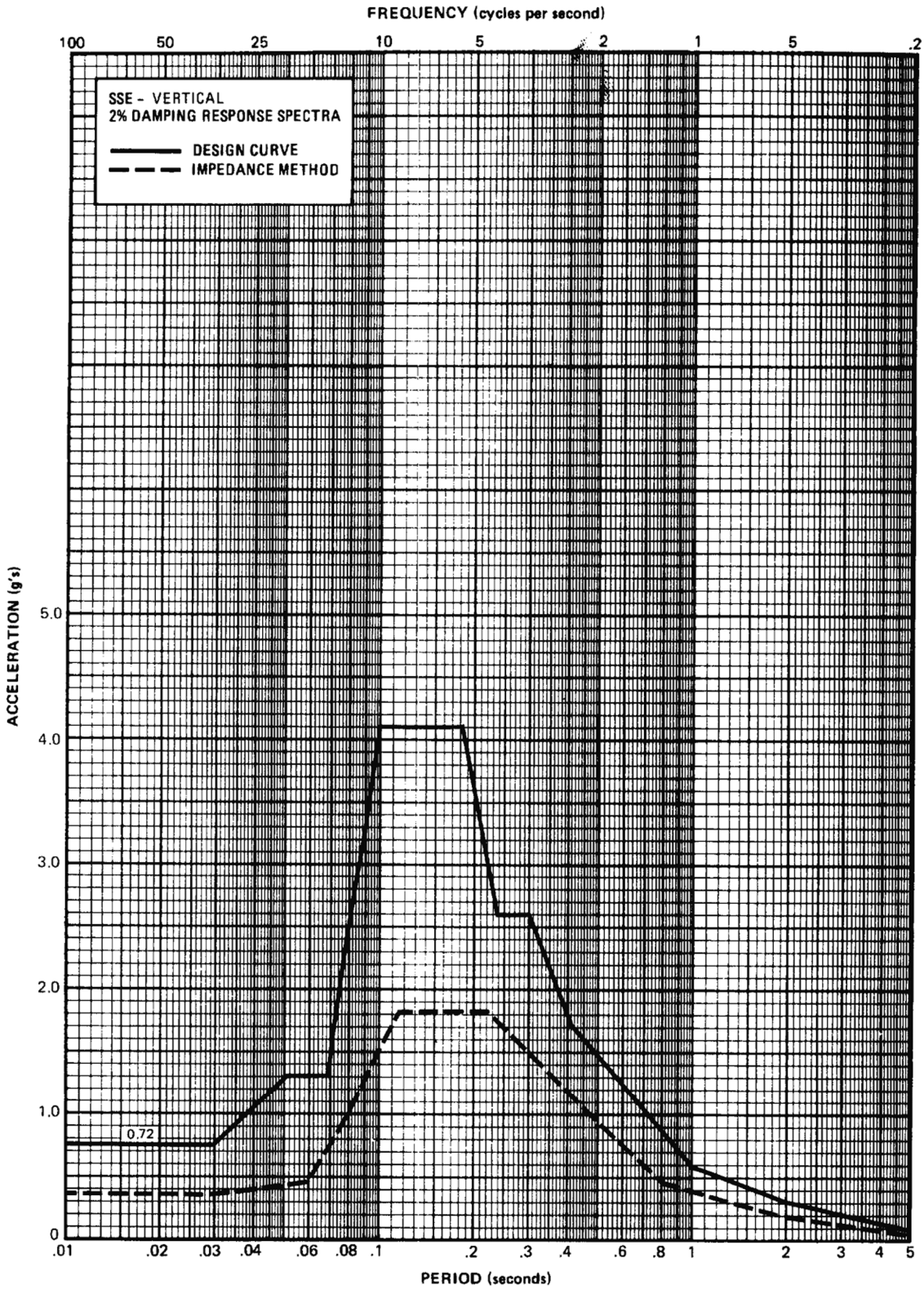

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| SOUTHERN COMPANY *Energy to Serve Your World®*<br>**VOGTLE**<br>**ELECTRIC GENERATING PLANT**<br>**UNIT 1 AND UNIT 2** | CONTROL BUILDING<br>EL. 260 FT. |
| | FIGURE 3D–146 |




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| | FIGURE 3D–147 |




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|---|---|
| SOUTHERN COMPANY *Energy to Serve Your World®*<br>**VOGTLE**<br>**ELECTRIC GENERATING PLANT**<br>**UNIT 1 AND UNIT 2** | CONTROL BUILDING<br>EL. 260 FT. |
| | FIGURE 3D–148 |

# APPENDIX 3E

## IMPEDANCE FUNCTIONS FOR AN ARBITRARILY SHAPED FOUNDATION ON A LAYERED MEDIUM

This appendix describes the procedure used to compute the impedance functions for an arbitrarily shaped foundation resting on a layered soil medium, for use in the soil-structure interaction analysis as specified in subsection 3.7.B.2.

The analytical techniques to obtain the impedance functions for flat rigid foundations of arbitrary shape placed on the surface of an elastic half space have been developed by Wong and Luco.[1] The equation of motion for the forced steady-state vibrations of an elastic half space excited by harmonic loads distributed over a region S of the plane surface (figure 3E-1) is:

$$(c^2 - \beta^2)\,\nabla(\nabla \cdot u) + \beta^2\nabla^2 u + \omega^2 u = 0 \qquad \chi_3 \geq 0 \tag{1}$$

in which u is the displacement vector $(u_1,u_2,u_3)e^{i\omega t}$ in the cartesian system of coordinates $(\chi_1,\chi_2,\chi_3)$ such that $\chi_3 = 0$ corresponds to the surface of the half space with $\chi_3 > 0$ representing the points within the half space. The symbols c and ß are the compressional and shear wave velocities, respectively.

Assuming that the surface tractions on the loaded region S are known, or equivalently, assuming that the stress components $\sigma_{j_3}$ (j = 1,2,3) on S are known, then a solution of equation (1) satisfying the mixed boundary-value problem on $\chi_3 = 0$, in which displacements are prescribed along the contact between the foundation and the soil while tractions are prescribed on the soil surface not covered by the foundation, is given by:

$$u_i(\chi_1,\chi_2,0) = - - \sum_{j=1}^{3} \int_S G_{ij}\,(\chi_i - \chi_1^0,\ \chi_2 - \chi_2^0)\,\sigma_{j3}\,(\chi_1^0,\ \chi_2^0,\ 0)\,d\chi_1^0\,d\chi_2^0 \tag{2}$$

for $\chi_3 = 0$. In equation 2, $G_{ij}(\chi_1^0 - \chi_1^0, \chi_2 - \chi_2^0, 0)$ denotes the ith displacement component at $(\chi_1,\chi_2,0)$ generated by a unit harmonic load acting at $(\chi_1^0,\chi_2^0,0)$.

To solve this integral equation 2 for an arbitrarily shaped foundation, the following numerical procedure is used:

A. The region S is divided into n rectangular subregions Sk (k = 1,2,...,n) as indicated in figure 3E-1.

B. The stress components $\sigma_{j3}$ are assumed to have constant values $\sigma_{j3}^{(k)}$ within each subregion $S_k$.

C. The boundary conditions are satisfied approximately by matching the average displacements within each subregion to the average value of the required compatible displacements.

Using the above approximations, integral equation 2 can be expressed in matrix equations as:

$$\begin{Bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ . \\ . \\ . \\ \bar{u}_n \end{Bmatrix} = \begin{bmatrix} [\phi]11\ [\phi]12.\ .\ .\ .\ [\phi]1n \\ \\ [\phi]21\ [\phi]22.\ .\ .\ .\ [\phi]2n \\ .\quad .\quad . \\ .\quad .\quad . \\ .\quad .\quad . \\ [\phi]n1\ [\phi]n2.\ .\ .\ .\ [\phi]nn \\ \\ \end{bmatrix} \begin{Bmatrix} \sigma_{j3}^{(1)} A_1 \\ \sigma_{j3}^{(2)} A_2 \\ . \\ . \\ . \\ \sigma_{j3}^{(n)} n \end{Bmatrix} \quad (3)$$

in which:

$\underset{\sim}{\bar{u}}_1 = \left\{ \bar{u}_{1i}, \bar{u}_{2i}, \bar{u}_{3i} \right\}^T$ are average displacements in subregion $S_i$.

$A_i$ = area if subregion $S_i$.

$[\phi]_{ij}$ = a 3x3 compliance submatrix relating the average displacements at subregion $S_i$ to the tractions at subregion $S_j$.

To calculate the compliance submatrices, four linear integrals on the Green's functions are performed for point loads at the surface of a layered stratum. The formula for $[\phi]_{ij}$ is:

$$[\phi]_{ij} = -\int_{Si} ds \int_{Sj} ds_o \left[ G(\omega, \underset{\sim}{r} - \underset{\sim o}{r}, \underset{\sim}{P}) \right] \quad (4)$$

in which [G] is the 3x3 Green's function matrix relating the displacement at the observation point $r = (\chi_1, \chi_2, 0)^T$ to a set of point loads at the source point $\underset{\sim o}{r} \cong \left(\chi_1^0, \chi_2^0, 0\right)^T$ • $\underset{\sim}{P}$ is the property vector associated with the underlying half space. By use of the average displacement matching approximation, the following symmetry exists even if $A_i \neq A_j$:

$$[\phi]_{ij}^T = [\phi]_{ji} \quad (5)$$

The property vector $\underset{\sim}{P}$ for a horizontally layered stratum can be characterized as:

$$\underset{\sim}{P} = (m, \mu_i, \beta_i, \rho_i, h_i, \nu_i, \varepsilon_i) \quad (6)$$

where:

- m = the number of layers.
- $\mu_i$ = the shear modulus of the ith layer.
- $\beta_i$ = the shear wave velocity of the ith layer.
- $\rho_i$ = the mass density of the ith layer.
- $H_i$ = the thickness of the ith layer.

$\nu_i$ = the Poisson's ratio of the ith layer.

$\xi_i$ = the critical damping coefficient of the ith layer.

The matrix [G] contains six independent elements. In order to reduce the number of independent variables and to render [G] dimensionless, a new matrix [G′] may be defined in the polar coordinate system {r, ψ, z}, as shown in figure 3E-2:

$$\left[G(\omega, \underset{\sim}{r} - \underset{\sim}{r_o}, \underset{\sim}{P})\right] = \frac{1}{\bar{\mu} r} \left[G'(b_0, \psi, \underset{\sim}{P'})\right] \quad (7)$$

where:

$$r = \left|\underset{\sim}{r} - \underset{\sim}{r_o}\right| = \sqrt{(\chi_1 - \chi_1^o)^2 + (\chi_2 - \chi_2^o)^2}$$

$$\psi = \arg\left(\underset{\sim}{r} - \underset{\sim}{r_o}\right) = \tan^{-1}\left(\frac{\chi_2 \chi_2^o}{\chi_1 - \chi_1^o}\right)$$

$$b_o = \frac{\omega r}{\bar{\beta}}$$

$\underset{\sim}{P'} = \{m, \mu_i', \beta_i', \rho_i', h_i', \nu_i, \varepsilon\}$ , the normalized property vector in which:

$$\mu_i' = \frac{\mu_i}{\bar{\mu}}$$

$$\beta_i' = \frac{\beta_i}{\bar{\beta}}$$

$$\rho_i' = \frac{\rho_i}{\bar{\rho}}$$

$$h_i' = \frac{\omega h_i}{\bar{\beta}}$$

The reference values of $\bar{\mu}$, $\bar{\beta}$ and $\bar{\rho}$, used to normalize $\underset{\sim}{P'}$ and [G´], are usually taken to be those of the top layer.

The dimensionless matrix [G´] is a function of four variables, $f_{rr}$, $f_{\psi r}$, $f_{rz}$, and $f_{zz}$, which are the Green's functions in the polar coordinate system.

The Green's functions for three-dimensional wave propagation in layered viscoelastic media have been formulated and solved by Luco and Apsel.[(2)(3)] In frequency domain, solution of the Green's functions in polar coordinates, which involve the Hankel transform-type integral representations of the displacement and stress components, can be expressed in the form:

$$I_n(b_o) = \int_0^\infty F(k, \omega, \underset{\sim}{P'}) J_n(kb_o)\, dk \quad (8)$$

for the concentrated point load applied at surface and displacement at the free surface observed at $b_o$ distance from the load point. The kernel F depends upon wave number k, frequency ω, and layer properties $\underset{\sim}{P'}$; whereas, the Bessel functions $J_n$ depend only upon $kb_o$.

The F integrands are evaluated in terms of factorizations of the upgoing and downgoing wave amplitudes in each layer. The semi-infinite integral in equation 8 can be reduced to the following finite integral:

$$I_n(b_o) = I_n(0) + \int_0^{k_\ell} [F(k,\omega,P') - F(k,0,P')]\, J_n(kb_o)\, dk \qquad (9)$$

in which the upper limit of integration, $k_\ell$, is defined by the convergence of the dynamic integrands to the static integrands, and $I_n(0)$ represents the static (= 0) integrals. Since the radial dependence $b_o$ appears only in the Bessel functions $J_n$, it is expedient to calculate the integrals; begin at $b_o = 0$ and end at $b_o = \frac{\omega r_{max}}{\beta}$ ($r_{max}$ is the maximum length of the foundation) in equally spaced intervals. This precalculated Green's function table can be repeatedly used in solving the compliance submatrices $[\phi]_{ij}$ in integral equation 4 using the Gaussian quadrature.

For a rigid foundation, the average displacements $\bar{u}_i$, evaluated at the center of subregion $S_i$ are given by:

$$\begin{aligned}
\bar{u}_{1i}(\chi_1^i,\chi_2^i,0) &= \Delta_1 - \theta_3 \chi_2^i \\
\bar{u}_{2i}(\chi_1^i,\chi_2^i,0) &= \Delta_2 + \theta_3 \chi_2^i \\
\bar{u}_{3i}(\chi_1^i,\chi_2^i,0) &= \Delta_3 + \theta_1 \chi_2^i - \theta_2 \chi_2^i
\end{aligned} \qquad (10)$$

where $\Delta_i$ (i = 1, 2, 3) corresponds to the amplitudes of the translational displacements at (0,0,0), while $\theta_i$ (i = 1,2,3), which is assumed to be small, corresponds to the amplitudes of the rotational displacements about the $\chi_i$ (i = 1, 2, 3) axes. From equation 3, the three corresponding traction components $\sigma_{j^3}^{(k)} A_k$ can be expressed in terms of $\bar{u}_i$, by inverting the matrix $[\phi]$. By substituting for $\bar{u}_i$ from equation 10, the surface tractions on the contact area may be expressed in terms of the translation $\Delta_1$ and rotation $\theta_i$ of the rigid foundation. Finally, the total harmonic load with components $(P_1,P_2,P_3)$ and the total harmonic moment with components $(M_1,M_2,M_3)$ acting on the contact area can be expressed in terms of traction components by means of the following relationships:

$$\begin{aligned}
P_1 &= \sum_{j=1}^{n} \sigma_{i3}^{(j)} Aj \qquad (i = 1,2,3) \\
M_1 &= \sum_{j=1}^{n} \chi_2^j \sigma_{33}^{(j)} \qquad (11) \\
M_2 &= \sum_{j=1}^{n} \chi_1^j \sigma_{33}^{(j)} \\
M_3 &= \sum_{j=1}^{n} \left[ \chi_1^j \sigma_{23}^{(j)} Aj - \chi_2^j \sigma_{13}^{(j)} Aj \right]
\end{aligned}$$

Substitution for contact tractions in terms of $\Delta_i$ and $\theta_i$ into equation 11 leads to the desired force-displacement relationship for the rigid foundation:

$$\begin{Bmatrix} P \\ M \end{Bmatrix} = [K] \begin{Bmatrix} \Delta \\ \theta \end{Bmatrix} \qquad (12)$$

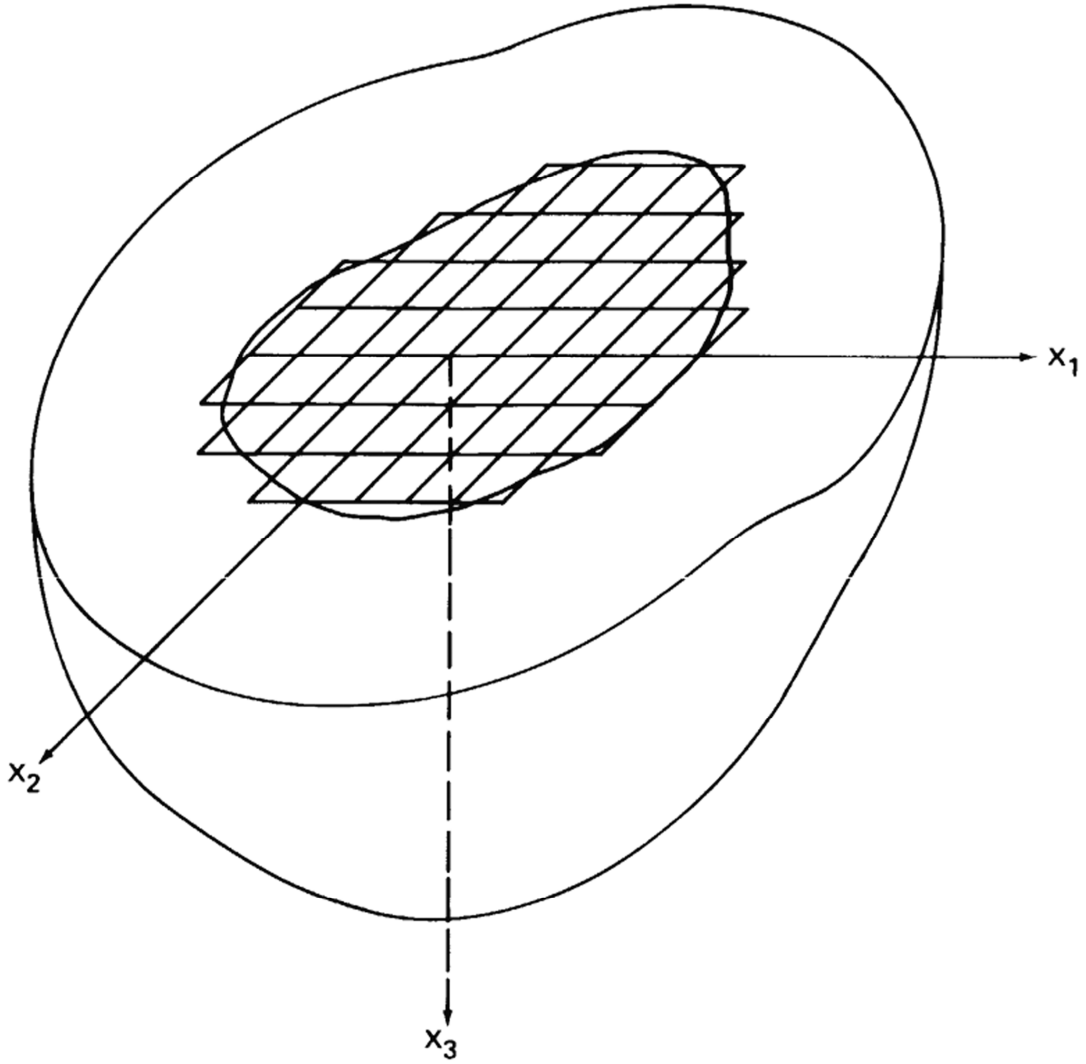
where [K] is the complex frequency dependent impedance functions for flat rigid foundations placed on the surface of an elastic half space.

## 3E.1 REFERENCES

1. Wong, H. L., and Luco, J. E., "Dynamic Response of Rigid Foundations of Arbitrary Shape," Earthquake Engineering and Structural Dynamics, Vol 4, pp 579-587, 1976.

2. Luco, J. E., and Apsel, R. J., "On the Green's Functions for the Layered Half Space," Part 1, Bulletin of the Seismological Society of America, Vol 73, pp 901-929, 1983.

3. Apsel, R. J., and Luco, J. E., "On the Green's Functions for the Layered Half Space," Part 2, Bulletin of the Seismological Society of America, Vol 73, pp 931-951, 1983.



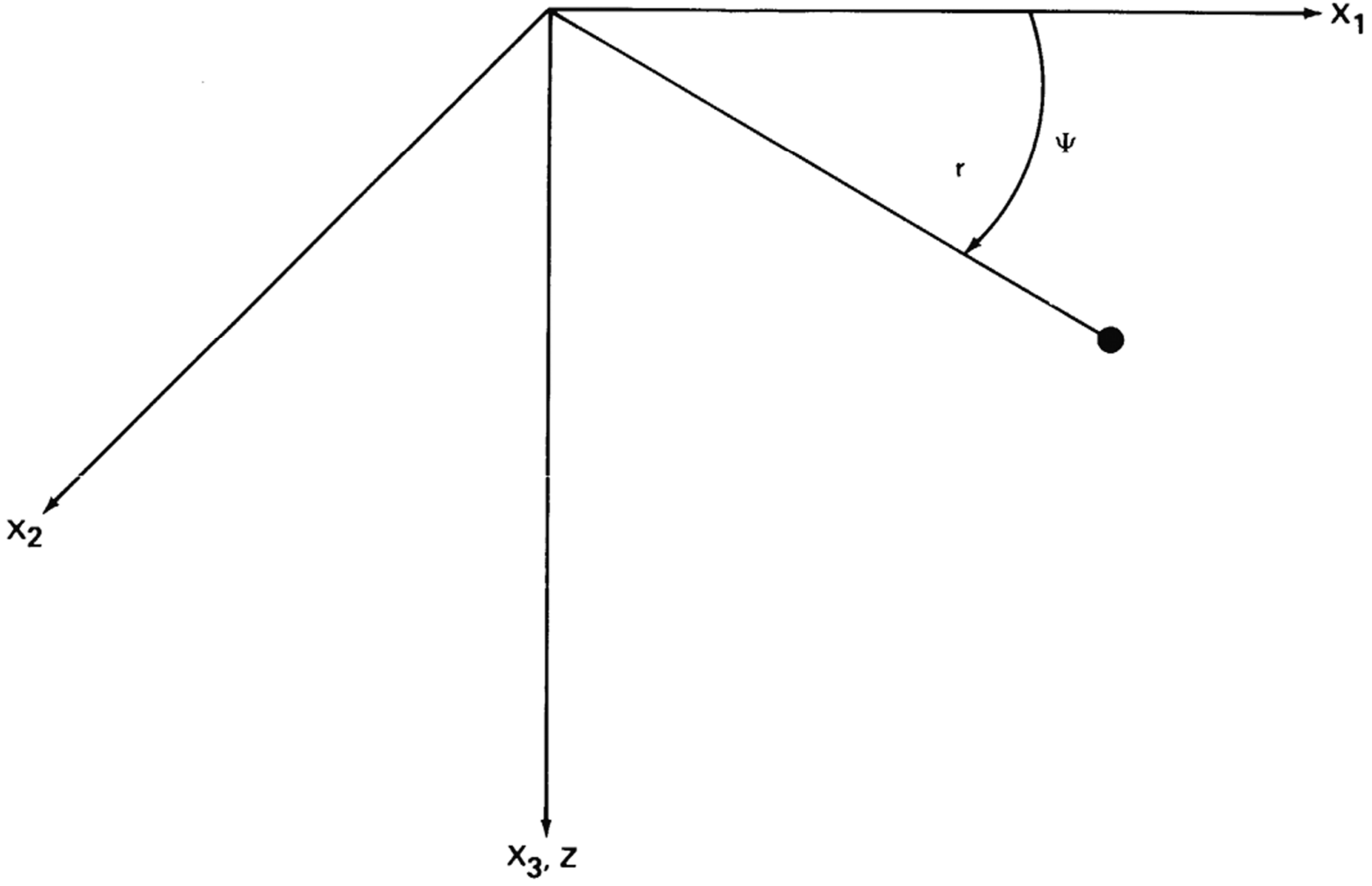

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MODEL AND COORDINATE SYSTEM

FIGURE 3E–1




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POLAR COORDINATE SYSTEM

FIGURE 3E–2

# APPENDIX 3F

# HAZARDS ANALYSIS

## 3F.1 INTRODUCTION

The VEGP power block has been designed to provide protection for safety-related equipment from hazards and events which could reasonably be expected to occur. This protection is provided to ensure that recovery from the event is possible, to ensure the integrity of the reactor coolant pressure boundary, to minimize the release of radioactivity, and to enable the plant to be placed in a safe condition.

This appendix provides the results of integrated hazards analyses for selected areas of the plant to demonstrate the type of analyses conducted for each safety-related area of the plant to ensure that the VEGP units can withstand the postulated events. A sample of the results of the analysis of level C and safety-related pump rooms on levels B and D of the auxiliary building as shown in table 3F-1 provides this example. Analyses are also provided for the pressure/temperature effects of a pipe rupture in the Unit 1 control building main steam line and main feedwater line isolation valve compartment, the flooding effects of pipe ruptures in the Unit 1 auxiliary and control building main steam line and main feedwater line isolation valve compartments, the effects of pipe ruptures in the Unit 1 auxiliary feedwater pump rooms, and the effects of a circulating water pipe rupture.

The items considered in the evaluation of each plant area include tornadoes, floods, missiles, pipe breaks, fires, and seismic events. (Refer to sections 3.3 through 3.7 and subsection 9.5.1.) Even though each area of the plant and each system are designed individually to properly consider the above events, an integrated analysis of rooms, systems, and events is performed to ensure that the above objectives are realized for each postulated event.

The hazards analyses are conducted on a room-by-room basis. All components within the room are reviewed for the effects of earthquake-induced failures, effects of high- and moderate-energy piping breaks (flooding, sprays, and jet impingement), and the effects of missiles.

The effects of the high-energy pipe breaks on equipment are reported in paragraph 3.6.2.5. Fire protection and the effects of fires in the various fire areas are discussed in subsection 9.5.1.

## 3F.2 ANALYSIS ASSUMPTIONS

In the analysis of an event or hazard, it is assumed that the plant is being operated in accordance with the requirements of the Technical Specifications. Should the event result in a turbine or reactor trip, the plant will be placed in a hot shutdown condition. If required by a limiting condition of operation or if recovery from the event will cause the plant to be shut down for an extended period of time, the plant will be taken to a cold shutdown condition. (Safe shutdown is discussed in section 7.4.)

During the hot shutdown condition, an adequate heat sink is provided to remove reactor core residual heat. Boration capability is provided to compensate for xenon decay and to maintain the required core shutdown margin. Boration is a long-term need because it is not required until approximately 25 h after shutdown.

Redundancy or diversity of systems and components is provided to enable continued operation at hot shutdown or to cool the reactor to a cold shutdown condition. If time is available, it is assumed that temporary repairs can be made to circumvent damages resulting from the hazard.

Loss of offsite power (LOSP) is not assumed, unless a trip of the turbine-generator system or the reactor protection system is a direct consequence of the hazard. All available systems, including nonsafety-related systems and those systems requiring operator action, may be employed to mitigate the consequences of the hazards.

In determining the availability of the systems required to mitigate the consequences of a hazard and those required to place the reactor in a safe condition, the direct consequences of the hazard are considered. The feasibility of carrying out operator actions is based on ample time and adequate access to the controls, motor control center, switchgear, etc., associated with the component required to accomplish the proposed action.

When the postulated hazard occurs in one of two or more redundant trains of a dual-purpose moderate-energy system, single failures of components in other trains (and associated supporting trains) are not assumed.

### 3F.2.1 EARTHQUAKE ANALYSIS ASSUMPTIONS

When evaluating the effects of any earthquake, no other major hazard or event is assumed, and no Seismic Category 1 equipment is assumed to fail as a result of the earthquake. Non-Seismic Category 1 equipment would not be available following the seismic event. Certain non-Seismic Category 1 components are designed and constructed to ensure that their failure could not reduce the functioning of a safe shutdown component to an unacceptable safety level. This criterion meets the intent of Regulatory Guide 1.29, Position C.2. Evaluation of component failure includes drop impact forces and secondary effects, such as spray and flooding from piping failures.

LOSP is assumed following a safe shutdown earthquake (SSE). An earthquake, as a single event, will affect the entire plant; hence, all the rooms dedicated to items associated with either safety-related train are considered in total.

### 3F.2.2 PIPE BREAK ANALYSIS ASSUMPTIONS

All high- and moderate-energy lines whose failure could reduce the functioning of a safe shutdown component to an unacceptable safety level are evaluated for pipe breaks or cracks. Thrust forces, jet impingement forces, and environmental effects are considered. Section 3.6 provides a description of the location and types of breaks and the forcing functions that are considered for analyzing pipe breaks.

Evaluation of environmental effects of moderate-energy pipe cracks has been made based on the characteristics of the flow from the postulated cracks. The locations of the cracks are discussed in paragraph 3.6.2.1. The evaluations include the effects of spraying or wetting on the safe shutdown equipment to assure that electrical safe shutdown equipment is not affected. The evaluation also includes the effect of flooding from the worst-case pipe crack in each room or general area. Flooding volumes are based on assuming automatic isolation or operator termination of flow to the pipe failure within a reasonable period after indication of the hazard. An interval of 30 min for operator's action after indication of flood is assumed.

### 3F.2.3 MISSILES ANALYSIS ASSUMPTIONS

There are two general sources of postulated internally generated missiles outside the containment, rotating component failure and pressurized component failure.

Section 3.5 provides a description of the design bases for the selection of missiles. Tables 3.5.1-1, 3.5.1-2, and 3.5.1-3 provide a listing of major missiles generated within the plant.

Analysis of impact from missiles is done for all rotating equipment and high-energy pressurized components.

### 3F.2.4 FLOODING ANALYSIS ASSUMPTIONS

In the event of a pipe failure, significant flooding might result and jeopardize the function of safety-related equipment required to mitigate the consequences of the pipe break or to maintain the plant in a safe shutdown condition.

Flooding rates are based on the worst-case pipe failure in each safety-related room or area. Through-wall cracks are postulated on moderate energy lines, seismic or non-seismic piping, to substantiate the effects due to spray and flooding on components which are required to mitigate the consequences of the event and/or safely shut down the plant; i.e., essential equipment. On high energy lines, terminal and intermediate breaks are postulated based on the lines' stress analyses. Full ruptures are assumed on all postulated breaks and are analyzed in part to determine flooding effects on essential equipment for the area in which the break is postulated.

The failure mode of non-Seismic Category 1 piping will be a critical crack rather than double-ended rupture, non-Seismic Category 1 piping in areas of safety-related structures where Seismic Category 2 over 1 interactions or flooding from a double-ended rupture could result in unacceptable interactions has been supported to withstand safe shutdown earthquake loads. The pipe support loads are determined by analyzing the piping system. For SSE loading, with the exception of the piping listed below, pipe stresses are also calculated to demonstrate they are maintained within faulted allowables:

- Duriron lines.
- Copper lines.
- ASTM A-120 galvanized lines.
- Air service lines.
- Process tubing $\leq$ 3/8 in.
- Instrument tubing.
- Moderate energy, non-Seismic Category 1 piping 2 in. and smaller in Unit 2 only.

For each area the worst flooding source is identified and analyzed for spray and/or flooding effects. The flooding rate is calculated based on the flooding sources reservoir capacity, piping dimensions, and fluid parameters. The level of the flood water is based on automatic isolation or operator action after a reasonable delay time following indication of flow from the breaks or crack. The delay time of 30 min for operator's action after indication of flood is assumed. If detection is not available, the full volume of the reservoir is assumed to flood the affected area. The maximum flood level is determined by distributing the flood volume over a surface area determined by room size and the size and geometry of large components within the room. Flow paths out of the room; i.e., open doors, grating, stairs, or large openings, are also considered.

The contents of the flooded area are then reviewed to ensure that no essential equipment is adversely affected. Additionally, flooding communication to adjacent rooms is evaluated to ensure that the event does not result in failure of essential components in adjacent rooms.

The analysis for flooding caused by failure of non-Seismic Category 1 tanks inside Category 1 buildings assumes each tank to completely rupture and empty into adjacent areas. The effects of open doors, stairs, and large openings are considered in the analysis. The tank's contents are then divided by the established boundaries' surface area to determine a flood level. The contents of the flooded area are then reviewed to ensure that no essential equipment is

adversely affected. The analysis considers the effects of the worst single active failure taken concurrent with the break. The effects of flooding from non-Seismic Category 1 tanks in the outside areas are discussed in paragraph 3.4.1.1.2.

Potential flooding due to probable maximum flood and probable maximum precipitation is discussed in section 3.4.

The equipment and floor drainage system is discussed in subsection 9.3.3. All water, released because of pipe breaks in the auxiliary building, drains to the common sump. Refer to paragraph 9.3.3.2.2.3 for a discussion of this design.

## 3F.3 PROTECTION MECHANISMS

The plant layout arrangement is based on maximizing the physical separation of redundant or diverse safety-related components and systems from each other and from nonsafety- related items. Therefore, if an accident occurs within the plant, there is minimal effect on other systems or components which are required for safe shutdown of the plant or to mitigate the consequence of the hazard.

Since it is not always feasible to provide separation in every hazard situation, other protection features are employed. These protection features include the following:

- Structural enclosures.
- Structural barriers.
- Restraints.
- Seismically designed components.
- Hardening.
- Orientation.

## 3F.4 HAZARDS EVALUATIONS

As stated previously, table 3F-1 provides a hazards evaluation of level C and safety-related pump rooms on levels B and D of the auxiliary building. Each room on those elevations is shown in drawings AX1D08A03-3, AX1D08A03-4, AX1D08A31, AX1D08A02-3, AX1D08A02-2 and AX1D08A04-4 and has been reviewed to ensure that the integrated design of the plant acceptably addresses all postulated hazards. Since the evaluations for equipment and components in all safety-related areas are documented in the project files and are available for audit, they are not provided in the FSAR.

Specific evaluations of certain areas of the plant have been of licensing concern in the past. These evaluations are provided in the following subsections.

### 3F.4.1 AUXILIARY FEEDWATER (AFW) PUMP ROOMS

The effects of a pipe break in the AFW pump rooms have been evaluated and the results of the Unit 1 analysis are presented in this appendix. The Unit 2 analysis is similar. The effects include room pressurization, temperature, flooding, and operability of the AFW system.

There are three separate AFW pump rooms, each housing one pump. Drawing 1X6DD300 provides plan and elevation views and nodal boundary of this area. Each of two motor-driven pumps is sized to deliver the feedwater flow required for decay heat removal. The single turbine-driven pump supplies twice the capacity of a motor-driven pump and is sufficient to remove

remove decay heat and, additionally, to cool down the reactor at a rate of up to 100°F/h not to exceed 100°F in any 1 hour period. The turbine-driven pump provides system diversity to both motor-driven pumps.

The results of the pressure/temperature analysis for the Unit 1 turbine-driven AFW pump room, as shown in table 3F-2, indicate that the maximum differential pressure on the walls will be less than 1.0 psi. The walls are capable of withstanding this pressure.

Analysis of AFW piping failures shows that loss of a redundant train does not prevent decay heat removal. The capability to provide adequate feedwater flow to remove decay heat is ensured by operation of either one of two motor-driven pumps or by operation of the turbine-driven pump.

Pressure-temperature response analysis for design basis main steam and main feedwater pipe breaks in the main steam tunnel has been performed. Design basis pipe breaks and locations are selected to identify maximum tunnel pressures to be accommodated by the AFW structure adjacent to the main steam tunnel.

Similarly, flooding caused by piping failures will not cause a loss of function of the AFW system because separation is provided between all three AFW pump rooms.

Watertight seals between these rooms prevent propagation of flooding and ensure that adequate capacity of the AFW system is maintained.

Analysis of the other hazards shows that adequate redundancy and separation are provided to ensure the operability of at least one train of the AFW system.

### 3F.4.2 MAIN STEAM ISOLATION VALVE (MSIV) AND MAIN FEEDWATER ISOLATION VALVE (MFIV) COMPARTMENT

The MSIV/MFIV compartment is located in the wing areas of the auxiliary building and the control building. Drawings 1X6DD301, 1X6DD302, AX6DD300, AX6DD301, and AX6DD302 provide plan and elevation views of this area. The main steam and main feedwater piping in this area consists of straight piping runs extending from the containment penetrations to torsional restraints mounted in the auxiliary building and control building walls through which these lines enter the main steam tunnel. The MSIVs, main steam safety valves, atmospheric relief valves, and MFIVs are in this compartment. Also in the compartment are branch piping lines of the AFW system, chemical addition system, steam supply to the turbine-driven AFW pump, bypass loops of the MSIVs, pressure instrumentation, and drains.

#### 3F.4.2.1 Break Size and Location

Main steam and feedwater piping in this compartment is designed to the criteria stated in paragraph 3.6.2.1 for those portions of the piping passing through the primary containment and extending to the first pipe whip restraint past the first outside isolation valve. In accordance with these criteria, no specific pipe breaks are postulated in the main run of these lines in the MSIV/MFIV compartment. However, to provide an additional level of assurance of operability of safety-related equipment in this compartment, the building structure and safety-related equipment are designed for the environmental conditions (pressure, temperature, and flooding) that would result from a break, equal in area to one cross-sectional pipe area, of either a main steam line or main feedwater line without steam generator tube bundle being uncovered. Main steam line breaks up to 1.0 $ft^2$ with steam generator tube bundle being uncovered have been considered for equipment qualification as discussed in paragraph 3.11.B.1.1.

Pressurization of the MSIV/MFIV compartment due to such a rupture is limited by providing adequate venting of the compartment and designing the compartment to withstand the maximum resultant pressure. Venting is accomplished by including adequate passageways between compartments or by other acceptable venting schemes. Engineered safety features required to bring the reactor to safe shutdown, which are located within these compartments, are designed to withstand the associated temperature, pressure, and humidity conditions.

The following cases are analyzed to determine the worst environmental conditions for the MSIV/MFIV compartment:

A. Case 1: Saturated steam blowdown from a main steam line break (MSLB) equivalent to the flow area of a single area rupture inside the restraint wall and a double area rupture outside the restraint wall. This case results in the maximum compartment pressure.
B. Case 2: Blowdown from a main feedwater line break equivalent to the flow area of a single area rupture. This case results in the maximum compartment flood level.
C. Case 3: Blowdown from MSLBs up to 1.0 $ft^2$ equivalent flow area with steam generator tube bundle being uncovered producing superheated blowdown. This case results in the maximum compartment temperatures and is discussed in paragraph 3.11.B.1.1.

### 3F.4.2.2 Method of Analysis

The case 1 analysis was performed using the Bechtel COPDA computer code, which is described in reference 1. The case 2 analysis was performed using the fluid flow equations identified in reference 2 for cold water flow. The case 3 analysis was performed using the GOTHIC computer code which is described in reference 4.

The MSIV area MSLB environmental conditions and the resulting impact on equipment qualification were evaluated using both the COPDA and GOTHIC computer codes. COPDA modeled the control and auxiliary sides of the MSIV/MFIV vault areas. Because reference 3 indicates the transient temperature for the auxiliary building MSIV/MFIV vault area is extremely similar to the transient for the same break in the control building, the detailed GOTHIC model was performed only on the control building MSIV/MFIV vault area. This area has compartments with smaller volumes and less flow area out of the break compartment than the auxiliary building MSIV/MFIV vault area (reference 3).

### 3F.4.2.3 Mass and Energy Release for Main Steam Line Break

For cases 1 and 2, blowdown mass and energy releases were calculated for a single area rupture inside the restraint wall and a double area rupture outside the restraint wall for both the auxiliary and control building MSIV/MFIV areas. The blowdown data for the double area rupture outside the restraint wall of the auxiliary building MSIV/MFIV area (Node 10) are presented in table 3F-3A.

For case 3, a spectrum of blowdown mass and energy releases was calculated for a spectrum of break sizes and power levels. The blowdown data for a 1.0 $ft^2$ single area rupture at 102% power (102% of 3579 MWt) inside the restraint wall in the control building MSIV/MFIV area are presented in table 3F-3B.

### 3F.4.2.4 Compartment Volumes and Vent Areas

For the Unit 1 analyses of the pressure temperature-transient following a MSLB (cases 1 and 3), the flow model of control volumes and intercompartment flow paths are illustrated in figures 3F-1 and 3F-2 and listed in tables 3F-3A and 3F-3B. The calculated Unit 1 flooding results from the case 2 analysis are shown in table 3F-3A. The calculated Unit 1 compartment pressure (case 1) and temperature (case 3) responses are listed in table 3F-3A and table 3F-3B. The Unit 2 analysis is similar.

### 3F.4.2.5 Initial Conditions

Tables 3F-3A and 3F-3B provide the Unit 1 initial conditions for case 1, case 2, and case 3 analyses.

### 3F.4.2.6 Design Provisions

The Plot of the time-history of the Unit 1 maximum break node compartment composite temperature for case 3 is given in figure 3F-3. The calculated Unit 1 flooding results from the case 2 analysis are shown in table 3F-3A.

Table 3F-3A provides the peak transient values of the Unit 1 compartment pressure analyses. Table 3F-3B provides the peak transient values of the Unit 1 compartment temperature analyses. The MSIV/MFIV compartment is designed to withstand these conditions. Essential safety-related equipment is qualified to these environmental conditions as discussed in section 3.11.

## 3F.4.3 EVALUATION OF REACTOR COOLANT SYSTEM (RCS) LOOP BRANCH LINE BREAKS

The evaluation of effects on safety-related equipment resulting from branch line breaks in the RCS is presented in table 3F-4. The evaluation shows that breaks in the RCS will not compromise the capability to safely shut down the plant.

## 3F.4.4 TURBINE BUILDING FLOODING EVALUATION

The flooding effects of a pipe break in the turbine building have been evaluated. Although the turbine building does not contain any essential safety-related equipment (however, see paragraph 7.2.1.1.2.F for trip instrumentation in the turbine building), it is connected to other safety-related structures by piping and electrical tunnels. However, it has been shown that the propagation of flood waters into the safety-related structures is precluded by design.

The design basis pipe failure in the turbine building is postulated to be a full circumferential break in the 96-in. circulating water piping to the condensers. A complete rupture of the condenser riser expansion joints has not been postulated as such joints have not been implemented in the VEGP design. The circulating water system design pressure is 80 psig, with a normal anticipated operating pressure of 45 psig. The maximum transient analysis pressure (unscheduled shutdown) is 54 psig. The flowrate through the break is conservatively estimated to be 612,000 gal/min, which is the combined runout flow of two circulating water pumps. This flowrate will cause the water level in the turbine building to rise at a rate of 0.72 ft/min.

If it is assumed that the turbine building flood detectors and the circulating water basin level detectors fail, the flooding could continue until the basin is empty. As the water level rises in one unit, the concrete block wall between units will fail due to high static water pressure, allowing the water to spread into both halves of the building. Calculations have shown that the block wall will

block wall will not withstand more than 8 ft of water and will fail before the water level reaches the operating deck at el 220 ft 0 in. If the entire circulating water system volume of $1.2 \times 10^6$ $ft^3$ is pumped into the turbine building, the resulting flood level in both units would be 211 ft.

Should flood water enter the main steam tunnels at the east or west end of the turbine building, it is precluded from entering the control and auxiliary buildings by sealed penetrations in the walls of the safety-related structures. The piping and electrical tunnels along the south wall at the centerline of the turbine building extend above grade level. Penetrations into the tunnels below grade are sealed.

A failure of the condensate feedwater system piping would result in significantly less flooding than a circulating water system failure. Even if the entire condensate/feedwater inventory flooded into the turbine building, the final flood level would be less than 1 ft above the level A floor.

## 3F.5 REFERENCES

1. Braddy, R.W. and Thiesing, J.W., Subcompartment Pressure and Temperature Transient Analysis, BN-TOP-4, Revision 0, Bechtel Corporation, San Francisco, California, July 1976.

2. Design for Pipe Break Effects, BN-TOP-2, Revision 2, Bechtel Corporation, San Francisco, California, June 1974.

3. Spryshak, J. J. and Iyengar, J., "Vogtle Electric Generating Plant Units 1 and 2. Main Steam Isolation Valve Area Equipment Thermal Response to Superheated Steam Releases," WCAP-13169, December 1991.

4. NAI 8907-02, Revision 8, “GOTHIC Containment Analysis Package User Manual.”

TABLE 3F-1 (SHEET 1 OF 84)

SAMPLE OF HAZARDS ANALYSIS RESULTS FOR AUXILIARY BUILDING (LEVELS B, C, AND D)

Room Number R-B15 Title Safety Injection Pump Room Train A

Flooding Analysis

- X Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G) will not preclude mitigation of the event or safe shutdown of the plant.
- X Flooding from sources external to the room is not credible even with a single active failure.
- ___ Other, see remarks.

Seismic Design Analysis

- ___ Only safety-related equipment (SRE) is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
- X The nonsafety-related equipment (NSRE) in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
- ___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
- ___ Other, see remarks.

Missile Analysis

- X No credible missile sources exist in the room.
- ___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
- ___ Other, see remarks.

Pipe Break Analysis

- ___ There are no high-energy lines in the room.
- ___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
- X Other, see remarks.

Remarks

1. The high-energy line breaks have been evaluated and will not preclude mitigation of the event or safe shutdown of the plant.

TABLE 3F-1 (SHEET 2 OF 84)

Listing of safety-related items in room R-B15

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Safety injection pump suction header isolation valve | HV-8807 A | | Y |
| 2. | Safety injection pump train A miniflow valve | HV-8814 | | Y |
| 3. | SIP A to RCS Cold Leg Isolation valve | HV-8821 A | | Y |
| 4. | Safety injection pump P6003 inlet valve | HV-8923 A | | Y |
| 5. | Safety injection pump room leak detection switch | LSH-9812 | | Y |
| 6. | Safety injection pump room leak detection switch | LSH-9816 | | Y |
| 7. | Charging pump header valve | HV-8924 | Y | Y |
| 8. | Cable tray | Train A | Y | Y |
| 9. | Safety injection pump | 1-1204-P6-003 | | Y |
| 10 | Air-handling units | 1-1555-A7-015-000 | | Y |
| 11. | Piping | 1-1204-003-1 1/2 in. | Y | Y |
| | | 1-1204-008-6 in. | Y | Y |
| | | 1-1202-117-3 in. | Y | Y |
| | | 1-1208-411-6 in. | Y | Y |
| | | 1-1204-011-6 in. | Y | Y |
| | | 1-1202-118-2 in. | Y | Y |
| | | 1-1202-116-2 in. | Y | Y |
| | | 1-1204-028-4 in. | | |
| | | 1-1204-014-4 in. | | |
| | | 1-1202-115-3 in. | | |
| | | 1-1202-113-2 in. | | |
| | | 1-1202-115-1 in. | Y | Y |
| | | 1-1202-117-1 in. | Y | Y |
| | | 1-1202-281-3/4 in. | Y | Y |
| | | 1-1202-297-1 in. | Y | Y |
| | | 1-1202-298-3/4 in. | Y | Y |
| | | 1-1202-372-2 in. | Y | Y |
| | | 1-1202-373-2 in. | Y | Y |
| | | 1-1202-475-2 in. | Y | Y |
| | | 1-1202-476-2 in. | Y | Y |
| | | 1-1204-008-3/4 in. | Y | Y |
| | | 1-1204-015-4 in. | Y | Y |
| | | 1-1204-028-3/4 in. | Y | Y |
| | | 1-1204-052-2 in. | Y | Y |
| | | 1-1204-128-3/4 in. | Y | Y |
| | | 1-1204-129-3/4 in. | Y | Y |
| | | 1-1204-130-3/4 in. | Y | Y |
| | | 1-1204-131-3/4 in. | Y | Y |
| | | 1-1204-280-1/2 in. | Y | Y |
| | | 1-1555-066-1 1/2 in. | Y | Y |
| | | 1-1609-031-3/4 in. | Y | Y |

TABLE 3F-1 (SHEET 3 OF 84)

Listing of safety-related items in room R-B15

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 12. | Safety injection pump discharge, flow transmitter | FT-0918 | | Y |
| 13. | Safety injection pump outlet, relief valve | PSV-8853A | | Y |
| 14. | Safety injection pump inlet, relief valve | PSV-8858 | | Y |
| 15. | Safety injection pump room cooler, high temperature switch | TISH-12210 | | Y |
| 16. | Train A raceway | Conduit and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 4 OF 84)

Room Number R-B18 Title Valve Gallery

Flooding Analysis

___ Flooding from sources within the room coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.
_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 5 OF 84)

Listing of safety-related items in room R-B18

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1202-162-3 in. | Y | Y |
| | | 1-1204-010-6 in. | Y | Y |
| | | 1-1204-004-3/4 in. | Y | Y |
| | | 1-1204-008-8 in. | Y | Y |

TABLE 3F-1 (SHEET 6 OF 84)

Room Number R-B19 Title Safety Injection Pump Room Train B

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

_X_ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.

___ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

Remarks

1. The high-energy line breaks have been evaluated and will not preclude mitigation of the event or safe shutdown of the plant.

TABLE 3F-1 (SHEET 7 OF 84)

Listing of safety-related items in room R-B19

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Safety injection pump suction header valve | HV-8807B | | Y |
| 2. | Safety injection pump miniflow isolation train B valve | HV-8813 | | Y |
| 3. | SIP B to RCS Cold Leg Isolation valve | HV-8821B | | Y |
| 4. | Safety injection pump inlet and miniflow isolation valve | HV-8920 | | Y |
| 5. | Safety injection pump P6004 inlet valve | HV-8923B | | Y |
| 6. | Safety injection pump room leak detection switch | LSH-9813 | | Y |
| 7. | Safety injection pump room leak detection switch | LSH-9817 | | Y |
| 8. | Cooler motor control switch | TISH 12216 | | Y |
| 9. | Train B raceway | Conduits and cabletrays | Y | Y |
| 10. | Safety injection pump | 1-1204-P6-004 | | Y |
| 11. | Air-handling unit | 1-1555-A7-016-000 | | Y |
| 12. | Piping | 1-1204-004-3 in. | Y | Y |
| | | 1-1202-381-2 in. | Y | Y |
| | | 1-1202-160-3 in. | Y | Y |
| | | 1-1202-162-3 in. | Y | Y |
| | | 1-1202-163-2 in. | Y | Y |
| | | 1-1204-004-1 1/2 in. | Y | Y |
| | | 1-1204-010-6 in. | Y | Y |
| | | 1-1204-012-8 in. | Y | Y |
| | | 1-1204-015-4 in. | Y | Y |
| | | 1-1202-161-2 in. | Y | Y |
| | | 1-1592-059-1 1/2 in. | Y | Y |
| | | 1-1592-050-1 1/2 in. | Y | Y |
| | | 1-1202-160-1 in. | Y | Y |
| | | 1-1202-289-3/4 in. | Y | Y |
| | | 1-1202-313-1 in. | Y | Y |
| | | 1-1202-380-2 in. | Y | Y |
| | | 1-1202-481-2 in. | Y | Y |
| | | 1-1204-002-3 in. | Y | Y |
| | | 1-1204-004-3/4 in. | Y | Y |
| | | 1-1204-004-1 1/2 in. | Y | Y |
| | | 1-1204-011-6 in. | Y | Y |
| | | 1-1204-015-3/4 in. | Y | Y |
| | | 1-1204-037-4 in. | Y | Y |
| | | 1-1204-280-1/2 in. | Y | Y |
| | | 1-1204-281-1/2 in. | Y | Y |
| | | 1-1204-314-3/4 in. | Y | Y |
| | | 1-1555-074-1 1/2 in. | Y | Y |
| | | 1-1609-030-3/4 in. | Y | Y |

TABLE 3F-1 (SHEET 8 OF 84)

Listing of safety-related items in room R-B19

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 13. | Safety injection pump discharge, flow transmitter | FT-0922 | | Y |
| 14. | Safety injection pump inlet, relief valve | PSV-8853B | | Y |
| 15. | Safety injection pump room cooler, high temperature switch. | TISH-12211 | | Y |

TABLE 3F-1 (SHEET 9 OF 84)

(Recycle Evaporator Abandoned in Place)

TABLE 3F-1 (SHEET 10 OF 84)

Listing of safety-related items in room R-C64, C65

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1228-174-in. | Y | Y |

TABLE 3F-1 (SHEET 11 OF 84)

(Waste Evaporator Abandoned in Place)

TABLE 3F-1 (SHEET 12 OF 84)

Listing of safety-related items in room R-C66, C67

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1228-168-2 in. | Y | Y |

TABLE 3F-1 (SHEET 13 OF 84)

Room Number R-C77 Title Boron Recycle Holdup Tank Room

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 14 OF 84)

Listing of safety-related items in room R-C77

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Boron recycle holdup tank | A-1210-T4-001 | | Y |
| 2. | Piping | 1-1208-151-3/4 in. | Y | Y |
| | | 1-1208-437-3/4 in. | Y | Y |
| | | 1-1208-493-3/4 in. | Y | Y |
| | | A-1561-102-10 in. | Y | Y |
| | | 1-1561-165-10 in. | Y | Y |
| | | | Y | Y |

TABLE 3F-1 (SHEET 15 OF 84)

Room Number R-C83 Title Corridor

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remark 1.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The SRE located in this room is not required for safe shutdown.

TABLE 3F-1 (SHEET 16 OF 84)

Listing of safety-related
items in room R-C83

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Recycle holdup tank/heating, ventilation, and air-conditioning (HVAC) vent valve | HV-12596 | N | Y |
| 2. | Recycle holdup tank/HVAC vent valve | HV-12597 | N | Y |
| 3. | Train A raceway | Conduits and cabletrays | N | Y |
| 4. | Train B raceway | Conduits and cabletrays | N | Y |
| 5. | Piping | 1-1208-017-2 in. | Y | Y |
| | | 2-1208-017-2 in. | Y | Y |
| | | 1-1208-020-2 in. | Y | Y |
| | | 1-1208-090-3 in. | Y | Y |
| | | 1-1208-093-3 in. | Y | Y |
| | | 1-1208-095-3 in. | Y | Y |
| | | 1-1208-340-3/8 in. | Y | Y |
| | | 1-1208-410-3 in. | Y | Y |
| | | 1-1208-437-3/8 in. | Y | Y |
| | | 1-1208-493-3/4 in. | Y | Y |
| | | 1-1213-026-3 in. | Y | Y |
| | | 1-1224-005-1 1/2 in. | Y | Y |
| | | 1-1224-006-4 in. | Y | Y |
| | | 1-1224-008-3 in. | Y | Y |
| | | 1-1224-011-2 in. | Y | Y |
| | | 1-1224-013-2 in. | Y | Y |
| | | 1-1224-014-2 in. | Y | Y |
| | | 1-1224-015-2 in. | Y | Y |
| | | 1-1224-016-2 in. | Y | Y |
| | | 1-1224-026-3 in. | Y | Y |
| | | 1-1224-027-3 in. | Y | Y |
| | | 1-1224-028-3 in. | Y | Y |
| | | 1-1224-066-3 in. | Y | Y |
| | | 1-1407-021-2 in. | Y | Y |

TABLE 3F-1 (SHEET 17 OF 84)

Room Number R-C88 Title Electrical Chase Train A

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The safe shutdown capability of the SRE located in this room is not impaired by this flood level.

TABLE 3F-1 (SHEET 18 OF 84)

Listing of safety-related
items in room R-C88

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train C Raceway | Conduits and cabletrays | Y | Y |
| 2. | Piping | 1-1202-104-8 in. | Y | Y |
| | | 1-1202-134-8 in. | Y | Y |

TABLE 3F-1 (SHEET 19 OF 84)

Room Number R-C89 Title Storage Room

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 20 OF 84)

Listing of safety-related items in room R-C89

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train D raceway | Conduits and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 21 OF 84)

Room Number R-C90 Title Residual Heat Removal (RHR) Exchanger Room Train A

Remarks

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.1.1.6.G), will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.

___ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 22 OF 84)

Listing of safety-related items in room R-C90

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A and B raceway | Conduits and cabletrays | Y | Y |
| 2. | RHR heat exchanger room leak detection switch | LSH 9874 | Y | Y |
| 3. | RHR miniflow valve | FV 0610 | Y | Y |
| 4. | RHR heat exchanger to pump valve | HV 8804A | Y | Y |
| 5. | RHR exchanger | 1-1205-E6-001 | Y | Y |
| 6. | Piping | 1-1205-007-8 in. | Y | Y |
| | | 1-1205-007-2 in. | Y | Y |
| | | 1-1205-013-3/4 in. | Y | Y |
| | | 1-1205-013-3 in. | Y | Y |
| | | 1-1208-411-8 in. | Y | Y |
| | | 1-1205-46-3/4 in. | Y | Y |
| | | 1-1205-005-8 in. | Y | Y |
| | | 1-1203-021-18 in. | Y | Y |
| | | 1-1203-021-14 in. | Y | Y |
| | | 1-1205-033-3/8 in. | Y | Y |
| | | 1-1203-122-1 in. | Y | Y |
| | | 1-1203-088-1 in. | Y | Y |
| | | 1-1203-086-1 in. | Y | Y |
| | | 1-1205-005-8 in. | Y | Y |
| | | 1-1205-008-8 in. | Y | Y |
| | | 1-1205-012-2 in. | Y | Y |
| | | 1-1205-014-3/4 in. | Y | Y |
| | | 1-1205-014-3 in. | Y | Y |
| | | 1-1205-022-3/4 in. | Y | Y |
| | | 1-1205-029-3/4 in. | Y | Y |
| | | 1-1205-030-2 in. | Y | Y |
| | | 1-1205-070-1/2 in. | Y | Y |
| 7. | HVAC ducts | Train A | Y | Y |
| 8. | Residual heat exchanger outlet valve | HV-0606 | Y | Y |
| 9. | Residual heat removal LP return bypass valve | FV-0618 | Y | Y |

TABLE 3F-1 (SHEET 23 OF 84)

Room Number R-C91 Title RHR Heat Exchanger Room Train B

Remarks

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.

___ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 24 OF 84)

Listing of safety-related items in room R-C91

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A and B raceways | Conduits and cabletrays | Y | Y |
| 2. | RHR pump to miniflow valve | FV 0611 | Y | Y |
| 3. | RHR heat exchanger room leak detection switch | LSH 9855 | Y | Y |
| 4. | RHR exchanger | 1-1205-E6-002 | Y | Y |
| 5. | Conduit | - | Y | Y |
| 6. | Piping | 1-1205-014-3/4 in. | Y | Y |
| | | 1-1205-014-3 in. | Y | Y |
| | | 1-1203-042-18 in. | Y | Y |
| | | 1-1203-042-14 in. | Y | Y |
| | | 1-1205-006-8 in. | Y | Y |
| | | 1-1204-012-8 in. | Y | Y |
| | | 1-1203-114-1 in. | Y | Y |
| | | 1-1203-116-1 in. | Y | Y |
| | | 1-1205-034-3/8 in. | Y | Y |
| | | 1-1205-045-3/4 in. | Y | Y |
| | | 1-1205-008-2 in. | Y | Y |
| | | 1-1205-008-8 in. | Y | Y |
| | | 1-1203-071-3/4 in. | Y | Y |
| | | 1-1203-075-3/4 in. | Y | Y |
| | | 1-1203-084-3/4 in. | Y | Y |
| | | 1-1203-111-3/4 in. | Y | Y |
| | | 1-1205-005-8 in. | Y | Y |
| | | 1-1205-007-8 in. | Y | Y |
| | | 1-1205-012-2 in. | Y | Y |
| | | 1-1205-018-1 in. | Y | Y |
| | | 1-1205-021-1 in. | Y | Y |
| | | 1-1205-029-3/4 in. | Y | Y |
| | | 1-1205-031-1 in. | Y | Y |
| | | 1-1205-032-1 in. | Y | Y |
| | | 1-1205-035-1 in. | Y | Y |
| | | 1-1205-036-1 in. | Y | Y |
| | | 1-1205-037-1 in. | Y | Y |
| | | 1-1205-038-1 in. | Y | Y |
| | | 1-1205-043-1 in. | Y | Y |
| | | 1-1205-044-1 in. | Y | Y |
| | | 1-1205-051-1 in. | Y | Y |
| | | 1-1205-052-1 in. | Y | Y |
| | | 1-1205-053-1 in. | Y | Y |
| | | 1-1205-054-1 in. | Y | Y |
| | | 1-1205-056-1 in. | Y | Y |
| | | 1-1205-057-1 in. | Y | Y |
| | | 1-1205-058-1 in. | Y | Y |
| | | 1-1205-059-1 in. | Y | Y |

TABLE 3F-1 (SHEET 25 OF 84)

Listing of safety-related items in room R-C91

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| | | 1-1205-072-1/2 in. | Y | Y |
| | | 1-1205-073-1/2 in. | Y | Y |
| | | 1-1205-074-1/2 in. | Y | Y |
| | | 1-1205-075-1/2 in. | Y | Y |
| 7. | HVAC ducts | Train B | Y | Y |
| 8. | Residual heat exchanger outlet valve | HV-0607 | Y | Y |
| 9. | Residual heat removal LP return bypass valve | FV-0619 | Y | Y |
| 10. | Residual heat removal exchanger train B to H injection pump valve | HV-8804B | Y | Y |

TABLE 3F-1 (SHEET 26 OF 84)

Room Number R-C95 Title Pipe Chase Train B

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remark 1.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.
_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 27 OF 84)

Listing of safety-related items in room R-C95

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train C raceway | Conduits and cableways | Y | Y |
| 2. | Piping | 1-1208-411-8 in. | Y | Y |
| | | 1-1204-012-8 in. | Y | Y |
| | | 1-1205-008-8 in. | Y | Y |
| | | 1-1206-006-8 in. | Y | Y |
| | | 1-1204-002-3 in. | Y | Y |
| | | 1-1204-037-4 in. | Y | Y |
| | | 1-1204-221-2 in. | Y | Y |
| | | 1-1208-003-3 in. | Y | Y |
| | | 1-1208-022-2 in. | Y | Y |
| | | 1-1208-095-3 in. | Y | Y |
| | | 1-1208-097-3 in. | Y | Y |
| | | 1-1208-103-3 in. | Y | Y |
| | | 1-1208-123-4 in. | Y | Y |
| | | 1-1208-131-2 in. | Y | Y |
| | | 1-1208-146-3 in. | Y | Y |
| | | 1-1208-410-3 in. | Y | Y |
| | | 1-1208-485-1 in. | Y | Y |
| | | 1-1208-493-3/4 in. | Y | Y |
| | | 1-1213-014-2 in. | Y | Y |
| | | 1-1213-017-2 in. | Y | Y |
| | | 1-1213-043-3 in. | Y | Y |
| | | 1-1407-007-4 in. | Y | Y |
| | | 1-1407-034-4 in. | Y | Y |
| | | 1-1561-103-10 in. | Y | Y |
| 3. | HVAC ducts | Train A | | Y |
| 4. | HVAC ducts | Train B | | Y |
| 5. | Temperature elements | TE-19723E | | Y |
| | | TE-19722E | | |

TABLE 3F-1 (SHEET 28 OF 84)

Room Number R-C99 Title Nonradioactive Pipe Chase

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

_X_ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.

___ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 29 OF 84)

Listing of safety-related items in room R-C99

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train B raceway | Conduits and cabletrays | Y | Y |
| 2. | Essential chilled water | 1-1592-020-2 1/2 in. piping | Y | Y |
| | | 1-1592-020-3 in. | Y | Y |
| | | 1-1592-031-2 1/2 in. | Y | Y |
| | | 1-1592-031-3 in. | Y | Y |
| | | 1-1592-052-1 1/2 in. | Y | Y |
| | | 1-1592-057-1 1/2 in. | Y | Y |
| | | 1-1592-044-2 1/2 in. | Y | Y |
| | | 1-1592-054-2 1/2 in. | Y | Y |
| | | 1-1592-026-2 in. | Y | Y |
| | | 1-1592-038-2 in. | Y | Y |
| | | A-1228-166-2 in. | Y | Y |
| | | 1-1228-166-2 in. | Y | Y |
| | | 1-1228-170-1 in. | Y | Y |
| | | 1-1228-174-2 in. | Y | Y |
| | | 1-1228-174-3 in. | Y | Y |
| | | A-1561-103-10 in. | Y | Y |
| | | 1-1592-044-2 in. | Y | Y |
| | | 1-1592-054-2 in. | Y | Y |

TABLE 3F-1 (SHEET 30 OF 84)

Room Number R-C99 Title Nonradioactive

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Safety-related function is not impaired by flooding or water spray.

TABLE 3F-1 (SHEET 31 OF 84)

Listing of safety-related items in room R-C103

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1205-003-14 in. | Y | Y |
| | | 1-1205-010-12 in. | Y | Y |
| | | 1-1205-007-8 in. | Y | Y |
| | | 1-1205-005-8 in. | Y | Y |
| 2. | Train A raceway | Conduits and cabletrays | Y | Y |
| 3. | HVAC ducts | Train A | | Y |

TABLE 3F-1 (SHEET 32 OF 84)

Room Number R-C104 Title Electrical Chase

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

_X_ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
___ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 33 OF 84)

Listing of safety-related items in room R-C104

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A and C raceway | Conduits and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 34 OF 84)

Room Number R-C105 Title Pipe Penetration Room Train A

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Safety-related function is not impaired by flooding or water spray.

TABLE 3F-1 (SHEET 35 OF 84)

Listing of safety-related items in room R-C105

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | HVAC ducts | Train A | | Y |
| 2. | Containment sump isolation valve | HV-8811A | | Y |
| 3. | Containment spray pump No. 1 suction | HV-9002A | | Y |
| 4. | Containment spray pump No. 2 suction | HV-9003A | | Y |
| 5. | Train D raceway | Conduits and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 36 OF 84)

Room Number R-C107 Title Vestibule

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Safety-related function is not impaired by flooding or water spray.

TABLE 3F-1 (SHEET 37 OF 84)

Listing of safety-related items in room R-C107

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | HVAC ducts | Train A | | Y |
| 2. | Temperature elements | TE-15212C | | Y |
| | | TE-15216C | | Y |
| 3. | Train D raceway | Conduits and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 38 OF 84)

Room Number R-C108 Title Steam Generator Blowdown Heat Exchanger Room

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remark 1.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
___ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
_X_ Other, see remark 2.

Missile Analysis

___ No credible missile sources exist in the room.
_X_ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
__ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.
_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Safety-related function is not impaired by flooding or water spray.
2. There is no essential equipment for safe shutdown located in this room; therefore, the postulated hazards have no effect on the safe shutdown capability.

TABLE 3F-1 (SHEET 39 OF 84)

Listing of safety-related items in room R-C108

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train C raceway | Conduits and cabletrays | Y | Y |
| 2. | Piping | 1-1206-001-12 in. | | Y |
| | | 1-1305-067-2 1/2 in. | Y | Y |
| | | 1-1305-068-2 1/2 in. | Y | Y |
| | | 1-1305-080-3 in. | Y | Y |
| | | 1-1305-080-4 in. | Y | Y |
| | | 1-1305-080-6 in. | Y | Y |
| | | 1-1305-098-3 in. | Y | Y |
| | | 1-1305-098-4 in. | Y | Y |
| | | 1-1305-099-3 in. | Y | Y |
| | | 1-1305-107-1 1/2 in. | Y | Y |
| | | 1-1305-107-3 in. | Y | Y |
| | | 1-1305-108-2 1/2 in. | Y | Y |
| | | 1-1305-109-2 1/2 in. | Y | Y |
| | | 1-1407-001-3 in. | Y | Y |
| | | 1-1407-002-3 in. | Y | Y |
| | | 1-1407-003-1 in. | Y | Y |
| | | 1-1407-003-3 in. | Y | Y |
| | | 1-1407-004-3 in. | Y | Y |
| | | 1-1407-005-1 in. | Y | Y |
| | | 1-1407-006-3 in. | Y | Y |
| | | 1-1407-015-3 in. | Y | Y |
| | | 1-1407-029-1 in. | Y | Y |
| | | 1-1407-035-4 in. | Y | Y |
| | | 1-1407-038-4 in. | Y | Y |
| | | 1-1407-039-4 in. | Y | Y |
| | | 1-1407-062-3 in. | Y | Y |
| | | 1-1407-066-3 in. | Y | Y |
| | | 1-1407-066-4 in. | Y | Y |
| | | 1-1407-067-3 in. | Y | Y |
| | | 1-1407-098-1 1/2 in. | Y | Y |
| | | 1-1407-124-3/4 in. | Y | Y |
| | | 1-1407-125-3/4 in. | Y | Y |
| | | 1-1407-126-3/4 in. | Y | Y |
| | | 1-1407-127-3/4 in. | Y | Y |
| | | 1-1407-128-3/4 in. | Y | Y |
| | | 1-1407-129-3/4 in. | Y | Y |
| | | 1-1407-130-3/4 in. | Y | Y |
| | | 1-1407-131-3/4 in. | Y | Y |
| | | 1-1407-132-3/4 in. | Y | Y |
| | | 1-1592-026-1 in. | Y | Y |
| | | 1-1592-038-1 in. | Y | Y |
| 3. | Temperature elements | TE-15212D | | Y |
| | | TE-15216D | | Y |

TABLE 3F-1 (SHEET 40 OF 84)

Room Number R-C109 Title Motor Control Center (MCC) Room

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

_X_ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
___ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Safety related function is not impaired by flooding or water spray.

TABLE 3F-1 (SHEET 41 OF 84)

Listing of safety-related items in room R-C109

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 42 OF 84)

Room Number R-C111

Title Chemical and Volume Control System (CVCS)
Normal Charging Pump Room

Remarks

Flooding Analysis

X Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G) will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

X The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

____The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

____Other, see remarks.

Missile Analysis

____No credible missile sources exist in the room.

X Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

____Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.

X The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 43 OF 84)

Listing of safety-related items in room R-C111

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | CVCS normal charging pump | 1-1208-P4-001 | N | Y |
| 2. | Piping | 1-1204-037-4 in. | Y | Y |
| | | 1-1204-168-3/4 in. | Y | Y |
| | | 1-1208-002-3 in. | Y | Y |
| | | 1-1208-003-2 in. | Y | Y |
| | | 1-1208-016-3 in. | Y | Y |
| | | 1-1208-123-4 in. | Y | Y |
| | | 1-1208-140-3/4 in. | Y | Y |
| | | 1-1208-142-3/4 in. | Y | Y |
| | | 1-1208-187-3/4 in. | Y | Y |
| | | 1-1208-189-3/4 in. | Y | Y |
| | | 1-1208-190-3/4 in. | Y | Y |
| | | 1-1208-280-1/2 in. | Y | Y |
| | | 1-1208-411-6 in. | Y | Y |
| | | 1-1208-440-2 in. | Y | Y |
| | | 1-1208-482-3/4 in. | Y | Y |
| | | 1-1208-483-3/4 in. | Y | Y |

TABLE 3F-1 (SHEET 44 OF 84)

Room Number R-C113 Title Boric Acid Batching Tank Room

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 45 OF 84)

Listing of safety-related
items in room R-C113

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cabletrays | Y | Y |
| 2. | Piping | 1-1202-122-1 1/2 in. | Y | Y |
| | | 1-1202-122-3 in. | Y | Y |
| | | 1-1202-125-2 in. | Y | Y |
| | | A-1208-232-3 in. | Y | Y |
| | | 1-1202-125-1 1/2 in | Y | Y |
| | | 1-1202-124-3 in. | Y | Y |
| | | 1/2-1208-241-3 in. | Y | Y |
| | | A-1208-034-3 in. | Y | Y |
| | | A-1208-230-1 in. | Y | Y |
| | | A-1208-230-3/4 in. | Y | Y |
| | | 1-1202-123-2 in. | Y | Y |
| | | 1-1202-124-1 1/2 in. | Y | Y |
| | | 1-1202-300-3/4 in. | Y | Y |
| | | 1-1202-473-2 in. | Y | Y |
| | | 1-1202-474-2 in. | Y | Y |
| | | 1-1204-177-8 in. | Y | Y |
| | | 1-1208-137-8 in. | Y | Y |
| | | 1-1208-241-3 in. | Y | Y |
| | | A-1208-492-3 in. | Y | Y |
| | | A-1228-166-2 in. | Y | Y |
| | | A-1228-166-1/2 in. | Y | Y |
| | | A-1228-228-3 in. | Y | Y |
| | | A-1609-034-3/4 in. | Y | Y |

TABLE 3F-1 (SHEET 46 OF 84)

Room Number R-C114 Title Valve Gallery

Remarks

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.

_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 47 OF 84)

Listing of safety-related items in room R-C114

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cabletrays | Y | Y |
| 2. | Refuel water tank to charging pump valve | LV 0112D | Y | Y |
| 3. | Piping | 1-1202-122-1 1/2 in. | Y | Y |
| | | 1-1202-123-2 in. | Y | Y |
| | | 1-1202-124-1 1/2 in. | Y | Y |
| | | 1-1202-125-2 in. | Y | Y |
| | | 1-1592-027-1 1/2 in. | Y | Y |
| | | 1-1592-037-1 1/2 in. | Y | Y |
| | | 1-1208-141-6 in. | Y | Y |
| | | 1-1208-144-4 in. | Y | Y |
| | | 1-1203-101-2 in. | Y | Y |
| | | 1-1203-144-4 in. | Y | Y |
| | | 1-1208-157-8 in. | Y | Y |
| | | 1-1208-095-3 in. | Y | Y |
| | | 1-1208-101-2 in. | Y | Y |
| | | 1-1208-137-8 in. | Y | Y |
| | | 1-1208-139-8 in. | Y | Y |
| | | 1-1208-147-3 in. | Y | Y |
| | | 1-1208-411-8 in. | Y | Y |
| | | 1-1208-485-1 in. | Y | Y |
| | | 1-1208-497-2 in. | Y | Y |
| | | 1-1208-498-2 in. | Y | Y |
| | | 1-1208-501-2 in. | Y | Y |
| 4. | Charging pump miniflow isolation valve | HV-8111A | Y | Y |
| 5. | Charging pump A suction valve | HV-8471A | Y | Y |
| 6. | Charging pump A discharge valve | HV-8485A | Y | Y |
| 7. | Charging pump miniflow isolation to RWST valve | HV-8508A, 8509B | Y | Y |
| 8. | Normal charging pump miniflow isolation valve | 1-1208-U4-150 | Y | Y |

TABLE 3F-1 (SHEET 48 OF 84)

Room Number R-C115 Title Centrifugal Charging Pump Room Train A

Remarks

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

___ No credible missile sources exist in the room.

_X_ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.

_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 49 OF 84)

Listing of safety-related items in room R-C115

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cabletrays | Y | Y |
| 2. | CVCS centrifugal charging pump | 1-1208-P6-002 | Y | Y |
| 3. | Charging pump room leak detection switch | LSH 9826 | | Y |
| 4. | Charging pump room leak detection switch | LSH 9830 | | Y |
| 5. | Air-handling unit | 1-1555A7-013 | Y | Y |
| 6. | Piping | 1-1202-122-1 1/2 in. | Y | Y |
| | | 1-1202-123-2 in. | Y | Y |
| | | 1-1202-124-1 1/2 in. | Y | Y |
| | | 1-1202-125-2 in. | Y | Y |
| | | 1-1202-370-2 in. | Y | Y |
| | | 1-1202-371-2 in. | Y | Y |
| | | 1-1208-141-6 in. | Y | Y |
| | | 1-1208-144-4 in. | Y | Y |
| | | 1-1592-027-1 1/2 in. | Y | Y |
| | | 1-1592-037-1 1/2 in. | Y | Y |
| | | 1-1204-012-8 in. | Y | Y |
| | | 1-1208-132-2 in. | Y | Y |
| | | 1-1208-191-1 in. | Y | Y |
| | | 1-1208-197-3/4 in. | Y | Y |
| | | 1-1208-198-1 in. | Y | Y |
| | | 1-1555-101-1 1/2 in. | Y | Y |
| | | 1-1555-134-1 1/2 in. | Y | Y |
| 7. | HVAC ducts | Train A | | Y |
| 8. | HVAC ducts | Train B | | Y |
| 9. | Charging pump room cooler temperature element | TE-12209 | | Y |

TABLE 3F-1 (SHEET 50 OF 84)

Room Number R-C116 Title Vestibule

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.

_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

Remarks

1. Safety-related equipment is not impaired by flooding or water spray.

TABLE 3F-1 (SHEET 51 OF 84)

Listing of safety-related items in room R-C116

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1202-003-4 in. | Y | Y |
| 2. | Train A and C raceway | 1-1202-004-4 in. | Y | Y |
| | | 1-1202-004-3 in. | Y | Y |
| | | 1-1202-003-3 in. | Y | Y |
| | | 1-1202-122-3 in. | Y | Y |
| | | 1-1202-124-3 in. | Y | Y |
| | | 1-1305-080-6 in. | Y | Y |
| | | Conduits and cabletrays | Y | Y |

TABLE 3F-1 (SHEET 52 OF 84)

Room Number R-C118 Title Centrifugal Charging Pump Train B

Remarks

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

_X_ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

___ No credible missile sources exist in the room.

_X_ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.

_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 53 OF 84)

Listing of safety-related items in room R-C118

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train B raceway | Conduits and cabletrays | Y | Y |
| 2. | CVCS centrifugal charging pump | 1-1208-P6-003 | Y | Y |
| 3. | Charging pump room leak detection switch | LSH 9827 | | Y |
| 4. | Charging pump room leak detection switch | LSH 9831 | | Y |
| 5. | Air-handling unit | 1-1555-A7-014 | Y | Y |
| 6. | Piping | 1-1202-167-1 1/2 in. | Y | Y |
| | | 1-1202-167-1 1/2 in. | Y | Y |
| | | 1-1202-168-2 in. | Y | Y |
| | | 1-1202-169-1 1/2 in. | Y | Y |
| | | 1-1202-170-2 in. | Y | Y |
| | | 1-1202-379-2 in. | Y | Y |
| | | 1-1202-378-2 in. | Y | Y |
| | | 1-1208-139-6 in. | Y | Y |
| | | 1-1208-145-4 in. | Y | Y |
| | | 1-1592-052-1 1/2 in. | Y | Y |
| | | 1-1592-067-1 1/2 in. | Y | Y |
| | | 1-1208-135-1 in. | Y | Y |
| | | 1-1208-200-3/4 in. | Y | Y |
| | | 1-1208-200-1 in. | Y | Y |
| | | 1-1208-201-3/4 in. | Y | Y |
| | | 1-1208-203-1 in. | Y | Y |
| | | 1-1208-204-3/4 in. | Y | Y |
| | | 1-1555-091-1 1/2 in. | Y | Y |
| | | 1-1555-135-1 1/2 in. | Y | Y |
| | | 1-1592-057-1 1/2 in. | Y | Y |
| 7. | HVAC ducts | Train A | | Y |
| 8. | Charging pump room cooler, high temperature switch and indicator | T1SH-12215 | | Y |

TABLE 3F-1 (SHEET 54 OF 84)

Room Number R-C119 Title Valve Gallery

Remarks

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

Pipe Break Analysis

___ There are no high-energy lines in the room.

_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

TABLE 3F-1 (SHEET 55 OF 84)

Listing of safety-related items in room R-C119

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train B raceway | Conduits and cabletrays | Y | Y |
| 2. | Charging pump train B, throttling valve | HV 1090B | Y | Y |
| | | HY-1090B | | |
| | | ZT-1090B | | |
| 3. | Piping | 1-1592-038-2 in. | Y | Y |
| | | 1-1592-026-2 in. | Y | Y |
| | | 1-1202-167-1 1/2 in. | Y | Y |
| | | 1-1202-168-2 in. | Y | Y |
| | | 1-1202-169-1 1/2 in. | Y | Y |
| | | 1-1202-170-2 in. | Y | Y |
| | | 1-1208-146-3 in. | Y | Y |
| | | 1-1208-099-3 in. | Y | Y |
| | | 1-1208-139-6 in. | Y | Y |
| | | 1-1208-145-4 in. | Y | Y |
| | | 1-1204-057-4 in. | Y | Y |
| | | 1-1208-099-2 in. | Y | Y |
| | | 1-1208-135-1 in. | Y | Y |
| | | 1-1208-137-8 in. | Y | Y |
| | | 1-1208-145-1 in. | Y | Y |
| | | 1-1208-303-2 in. | Y | Y |
| | | 1-1208-499-2 in. | Y | Y |
| | | 1-1592-052-1 1/2 in. | Y | Y |
| | | 1-1592-057-1 1/2 in. | Y | Y |
| 4. | Charging pump miniflow isolation valve | HV-8111B | | Y |
| 5. | Charging pump discharge, train B valve | HV-8438, HV-8485V | Y | Y |
| 6. | Charging pump suction, train B valve | HV-8471B | Y | Y |
| 7. | Charging pump miniflow isolation to RWST valves | HV-8508B, HV-8509B | | |

TABLE 3F-1 (SHEET 56 OF 84)

Room Number R-C120 Title Vestibule

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.F), will not preclude mitigation of the event or safe shutdown of the plant.
__ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 57 OF 84)

Listing of safety-related items in room R-C120

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train B raceway | Conduit and cabletrays | Y | Y |
| 2. | Piping | 1-1592-038-2 in. | Y | Y |
| | | 1-1592-026-2 in. | Y | Y |
| | | 1-1592-052-1 1/2 in. | Y | Y |
| | | 1-1592-057-1 1/2 in. | Y | Y |
| | | 1-1202-167-1 1/2 in. | Y | Y |
| | | 1-1202-168-2 in. | Y | Y |
| | | 1-1202-169-1 1/2 in. | Y | Y |
| | | 1-1202-170-2 in. | Y | Y |
| | | 1-1202-151-4 in. | Y | Y |
| | | 1-1202-169-3 in. | Y | Y |
| | | 1-1202-151-3 in. | Y | Y |
| | | 1-1202-006-3 in. | Y | Y |
| | | 1-1202-006-4 in. | Y | Y |
| | | 1-1202-151-4 in. | Y | Y |
| | | 1-1202-167-3 in. | Y | Y |
| | | 1-1202-290-3/4 in. | Y | Y |
| | | 1-1202-316-3/4 in. | Y | Y |
| | | 1-1202-479-2 in. | Y | Y |
| | | 1-1202-480-2 in. | Y | Y |
| | | 1-1228-166-2 in. | Y | Y |
| | | 1-1609-030-3/4 in. | Y | Y |
| | | 1-1609-031-3/4 in. | Y | Y |
| | | 1-1609-032-3/4 in. | Y | Y |
| | | 1-1609-033-3/4 in. | Y | Y |
| | | 1-1609-034-3/4 in. | Y | Y |
| 3. | Refueling water tank to charging pump valve | LV-0112E | | |

TABLE 3F-1 (SHEET 58 OF 84)

Room Number R-C131 Title Vestibule

Flooding Analysis

__ Flooding from sources within the room coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
__ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remark 1.

Seismic Design Analysis

__ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
__ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
__ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
__ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
__ Other, see remarks.

Pipe Break Analysis

__ There are no high-energy lines in the room.
_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.
__ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown, capability of the SRE.

TABLE 3F-1 (SHEET 59 OF 84)

Listing of safety-related items in room R-C131

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1205-003-12 in. | Y | Y |
| | | 1-1407-001-3 in. | Y | Y |
| | | 1-1407-002-3 in. | Y | Y |
| | | 1-1407-003-3 in. | Y | Y |
| | | 1-1407-004-3 in. | Y | Y |

TABLE 3F-1 (SHEET 60 OF 84)

Room Number R-C133 Title Pipe Penetration Room Train A

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 61 OF 84)

Listing of safety-related items in room R-C133

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cabletrays | Y | Y |
| 2. | Piping | 1-1205-003-14 in. | Y | Y |
| | | 1-1205-003-12 in. | Y | Y |
| | | 1-1205-010-12 in. | Y | Y |
| | | 1-1205-007-8 in. | Y | Y |
| | | 1-1206-005-8 in. | Y | Y |
| | | 1-1205-028-1 in. | Y | Y |
| | | 1-1205-066-1 1/2 in. | Y | Y |
| 3. | HVAC ducts | Train A | | Y |

TABLE 3F-1 (SHEET 62 OF 84)

Room Number R-C134 Title Pipe Penetration Room Train A

Flooding Analysis
___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis
___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis
_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis
_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks
1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 63 OF 84)

Listing of safety-related items in room R-C134

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Containment sump isolation valve | HV-8811A | | Y |
| 2. | RHR encapsulation vessel | V-1-1205 V4 001 | Y | Y |
| 3. | Containment spray pump emergency sump isolation valve | HV-9002A | | Y |
| 4. | Containment spray pump suction valve | HV-9003A | | Y |
| 5. | Piping | 1-1205-028-14 in. | Y | Y |
| | | 1-1205-061-20 in. | Y | Y |
| | | 1-1205-041-14 in. | Y | Y |
| | | 1-1206-001-12 in. | Y | Y |
| | | 1-1206-061-18 in. | Y | Y |
| | | 1-1205-003-12 in. | Y | Y |
| | | 1-1205-062-3/4 in. | Y | Y |
| | | 1-1205-063-3/4 in. | Y | Y |
| | | 1-1206-001-1 in. | Y | Y |
| | | 1-1206-076-3/4 in. | Y | Y |
| | | 1-1206-077-3/4 in. | Y | Y |

TABLE 3F-1 (SHEET 64 OF 84)

Room Number UC-C14 Title Pipe Chase

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remark 1.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

___ No credible missile sources exist in the room.
_X_ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

___ There are no high-energy lines in the room.
_X_ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 65 OF 84)

Listing of safety-related items in room UC-C14

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Piping | 1-1204-002-3 in. | Y | Y |
| | | 1-1208-146-3 in. | Y | Y |
| | | 1-1208-103-3 in. | Y | Y |
| | | 1-1208-097-3 in. | Y | Y |
| | | 1-1208-095-3 in. | Y | Y |
| | | 1-1208-411-8 in. | Y | Y |
| | | 1-1208-123-4 in. | Y | Y |
| | | 1-1208-410-3 in. | Y | Y |
| | | 1-1208-107-1 in. | Y | Y |
| | | 1-1208-022-2 in. | Y | Y |
| | | 1-1208-003-3 in. | Y | Y |
| | | 1-1204-037-4 in. | Y | Y |
| | | 1-1204-012-8 in. | Y | Y |
| | | 1-1208-485-1 in. | Y | Y |

TABLE 3F-1 (SHEET 66 OF 84)

Room Number R-D48 Title RHR Pump Room Train A

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

_X_ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remark 1.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

__ Other, see remark 2.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.

___ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

Remarks

1. The SRE will not be adversely affected by the maximum design flood.

TABLE 3F-1 (SHEET 67 OF 84)

Listing of safety-related items in room R-D48

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Leak detection switch | LSH 9856 | | Y |
| 2. | Leak detection switch | LSH 9860 | | Y |
| 3. | RHR pump 1 inlet valve | HV 8812A | Y | Y |
| 4. | RHR 1 cold leg isolation valve | HV 8716A | Y | Y |
| 5. | Residual heat removal pump cooler temperature element | TE-12206 | Y | Y |
| 6. | Train A raceway | Conduits and cabletrays | Y | Y |
| 7. | RHR pump | 1-1205-P6-001 | Y | Y |
| 8. | Piping | 1-1205-019-2 in. | Y | Y |
| | | 1-1205-023-2 in. | Y | Y |
| | | 1-1205-039-12 in. | Y | Y |
| | | 1-1205-013-3 in. | Y | Y |
| | | 1-1202-110-2 in. | Y | Y |
| | | 1-1202-111-2 in. | Y | Y |
| | | 1-1203-088-1 in. | Y | Y |
| | | 1-1203-086-1 in. | Y | Y |
| | | 1-1205-007-8 in. | Y | Y |
| | | 1-1205-003-14 in. | Y | Y |
| | | 1-1205-009-8 in. | Y | Y |
| | | 1-1205-005-8 in. | Y | Y |
| | | 1-1205-005-2 in. | Y | Y |
| | | 1-1204-001-12 in. | Y | Y |
| | | 1-1205-016-2 in. | Y | Y |
| | | 1-1205-039-2 in. | Y | Y |
| | | 1-1205-070-1/2 in. | Y | Y |
| | | 1-1205-071-1/2 in. | Y | Y |
| | | 1-1205-192-8 in. | Y | Y |
| | | 1-1202-478-2 in. | Y | Y |

TABLE 3F-1 (SHEET 68 OF 84)

Room Number R-D76 Title Containment Spray Pump Room Train A

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
_X_ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE

TABLE 3F-1 (SHEET 69 OF 84)

Listing of safety-related items in room R-D76

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Leak detection switch | LSH 9872 | | Y |
| 2. | Refueling water storage tank to containment spray pump 1, valve | HV 9017A | | Y |
| 3. | Leak detection switch | LSH 9868A | | Y |
| 4. | Train A raceway | Conduits and cabletrays | Y | Y |
| 5. | Containment spray pump | 1-1206-P6-001 | Y | Y |
| 6. | Piping | 1-1206-001-12 in. | Y | Y |
| | | 1-1206-005-8 in. | Y | Y |
| | | 1-1202-110-2 in. | Y | Y |
| | | 1-1202-111-2 in. | Y | Y |
| | | 1-1206-036-3 in. | Y | Y |
| | | 1-1206-003-10 in. | Y | Y |
| | | 1-1202-003-2 in. | Y | Y |
| | | 1-1202-003-3 in. | Y | Y |
| | | 1-1204-007-10 in. | Y | Y |
| | | 1-1202-004-2 in. | Y | Y |
| | | 1-1202-004-3 in. | Y | Y |
| | | 1-1202-105-2 in. | Y | Y |
| | | 1-1202-294-3/4 in. | Y | Y |
| | | 1-1202-477-2 in. | Y | Y |
| | | 1-1206-005-3 in. | Y | Y |
| | | 1-1206-022-3/4 in. | Y | Y |
| | | 1-1206-023-3/4 in. | Y | Y |
| | | 1-1206-031-3/4 in. | Y | Y |
| | | 1-1206-047-3 in. | Y | Y |
| | | 1-1206-051-3/4 in. | Y | Y |
| | | 1-1206-070-1 in. | Y | Y |
| | | 1-1206-152-2 in. | Y | Y |
| | | 1-1206-223-2 in. | Y | Y |
| 7. | Containment sump pump relief valve | PSV-9007A | | Y |
| 8. | Containment spray pump room cooler temperature switch | TISH-12207 | Y | Y |
| 9. | Containment spray train A | 1-1206-6001 | | |

TABLE 3F-1 (SHEET 70 OF 84)

Room Number R-D77 Title Containment Spray Pump Room Train B

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G) will not preclude mitigation of the event or safe shutdown of the plant.
_X_ Flooding from sources external to the room is not credible even with a single active failure.
___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

___ No credible missile sources exist in the room.
_X_ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The only equipment in this room is associated with containment spray system, which is not essential in this case.

TABLE 3F-1 (SHEET 71 OF 84)

Listing of safety-related
items in room R-D77

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Leak detection switch | LSH 9869 | | Y |
| 2. | Radwaste to containment spray pump 2, valve | HV 9017B | | Y |
| 3. | Train B raceway | Conduits and cabletrays | Y | Y |
| 4. | Containment spray pump | 1-1206-P6-002 | | Y |
| 5. | Air-handling unit | 1-1555-A7-010 | | Y |
| 6. | Piping | 1-1202-151-2 in. | Y | Y |
| | | 1-1202-006-2 in. | Y | Y |
| | | 1-1204-038-10 in. | Y | Y |
| | | 1-1206-004-12 in. | Y | Y |
| | | 1-1206-002-12 in. | Y | Y |
| | | 1-1206-006-8 in. | Y | Y |
| | | 1-1592-044-1 1/2 in. | Y | Y |
| | | 1-1592-054-1 1/2 in. | Y | Y |
| | | 1-1202-006-3 in. | Y | Y |
| | | 1-1202-310-3/4 in. | Y | Y |
| | | 1-1202-483-2 in. | Y | Y |
| | | 1-1204-221-2 in. | Y | Y |
| | | 1-1206-004-10 in. | Y | Y |
| | | 1-1206-006-2 in. | Y | Y |
| | | 1-1206-019-3 in. | Y | Y |
| | | 1-1206-027-3/4 in. | Y | Y |
| | | 1-1206-048-3 in. | Y | Y |
| | | 1-1206-052-3/4 in. | Y | Y |
| | | 1-1206-071-1 in. | Y | Y |
| | | 1-1555-070-1/2 in. | Y | Y |
| | | 1-1609-035-3/4 in. | Y | Y |
| | | 1-1609-036-3/4 in. | Y | Y |
| 7. | Containment spray pumproom cooler temperature switch | TISH-12207 | Y | Y |
| 8. | Containment spray pumproom, train B level switch | LSH-9873 | Y | Y |
| 9. | Containment spray pumproom cooler temperature switch | TISH-12213 | Y | Y |

TABLE 3F-1 (SHEET 72 OF 84)

Room Number R-D100 Title Pipe Chase Train A

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

X No credible missile sources exist in the room.
___ Missiles from rotating component generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 73 OF 84)

Listing of safety-related
items in room R-D100

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Pipe chase leak detection switch | LSH 9842 | | Y |
| 2. | Pipe chase leak detection switch | LSH 9846 | | Y |
| 3. | Train A raceway | Conduits and cabletrays | Y | Y |
| 4. | Piping | 1-1205-007-8 in. | Y | Y |
| | | 1-1592-031-2 1/2 in. | Y | Y |
| | | 1-1592-020-2 1/2 in. | Y | Y |
| | | 1-1205-003-14 in. | Y | Y |
| | | 1-1204-001-12 in. | | Y |
| | | 1-1202-111-2 in. | Y | Y |
| | | 1-1202-110-2 in. | Y | Y |
| | | 1-1205-010-12 in. | Y | Y |
| | | 1-1205-009-8 in. | Y | Y |
| | | 1-1204-006-24 in. | Y | Y |
| | | 1-1204-192-8 in. | Y | Y |

TABLE 3F-1 (SHEET 74 OF 84)

Room Number R-D101 Title Pipe Chase Train B

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 75 OF 84)

Listing of safety-related items in room R-D101

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train B raceway | Conduits and cabletrays | Y | Y |
| 2. | Piping | 1-1205-004-14 in. | Y | Y |
| | | 1-1205-008-8 in. | Y | Y |
| | | 1-1206-006-8 in. | | Y |
| | | 1-1206-002-12 in. | | Y |
| | | 1-1202-155-2 in. | Y | Y |
| | | 1-1202-156-2 in. | Y | Y |
| | | 1-1205-009-8 in. | Y | Y |
| | | 1-1204-006-12 in. | Y | Y |
| | | 1-1204-038-14 in. | Y | Y |
| | | 1-1204-006-24 in. | Y | Y |
| | | 1-1208-131-2 in. | Y | Y |
| | | 1-1208-241-3 in. | Y | Y |
| | | 1-1208-242-2 in. | Y | Y |
| | | 1-1204-221-2 in. | Y | Y |
| | | 1-1205-105-2 in. | Y | Y |
| | | 1-1208-017-2 in. | Y | Y |
| | | 1-1208-241-3 in. | Y | Y |
| | | 1-1208-450-2 in. | Y | Y |

TABLE 3F-1 (SHEET 76 OF 84)

Room Number R-D121 Title Passage

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
_X_ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 77 OF 84)

Listing of safety-related items in room R-D121

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | RHR miniflow switch | FIS-0610 | Y | Y |
| 2. | Train A raceway | Conduits and cabletrays | Y | Y |
| 3. | Residual heat removal LP Yet bypass, train A flow transmitter | FT-0618 | Y | Y |

TABLE 3F-1 (SHEET 78 OF 84)

Room Number R-D122 Title Passage

Flooding Analysis

___ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
_X_ Flooding from sources external to the room is not credible even with a single active failure.
___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

_X_ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
___ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. The flooding from sources located in this room will not impair the safe shutdown capability of the SRE.

TABLE 3F-1 (SHEET 79 OF 84)

Listing of safety-related items in room R-D122

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cabletrays | Y | Y |
| 2. | Piping | 1-1205-026-3/4 in. | Y | Y |
| | | 1-1555-061-1 1/2 in. | Y | Y |
| | | 1-1592-020-2 1/2 in. | Y | Y |
| | | 1-1592-028-1 1/2 in. | Y | Y |
| | | 1-1592-031-2 1/2 in. | Y | Y |
| | | 1-1592-036-1 1/2 in. | Y | Y |
| | | 1-1592-053-1 1/2 in. | Y | Y |
| | | 1-1592-056 1 1/2 in. | Y | Y |

TABLE 3F-1 (SHEET 80 OF 84)

Room Number R-D128 Title Cooler Room Train A

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G) will not preclude mitigation of the event or safe shutdown of the plant.

___ Flooding from sources external to the room is not credible even with a single active failure.

___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).

_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.

___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.

___ Other, see remarks.

Missile Analysis

___ No credible missile sources exist in the room.

___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).

_X_ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.

___ The high-energy line breaks have been evaluated and do not adversely affect SRE.

___ Other, see remarks.

Remarks

1. Missiles from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G) will not preclude mitigation of the event or safe shutdown of the plant.

TABLE 3F-1 (SHEET 81 OF 84)

Listing of safety-related items in room R-D128

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train A raceway | Conduits and cableways | Y | Y |
| 2. | Piping | 1-1592-036-1 1/2 in. | Y | Y |
| | | 1-1592-028-1 1/2 in. | Y | Y |
| | | 1-1592-020-2 1/2 in. | Y | Y |
| | | 1-1592-031-2 1/2 in. | Y | Y |
| 3. | Air-handling unit | 1-1555-A7-007 | Y | Y |

TABLE 3F-1 (SHEET 82 OF 84)

Room Number R-D130 Title Cooler Room Train B

Flooding Analysis

_X_ Flooding from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.
___ Flooding from sources external to the room is not credible even with a single active failure.
___ Other, see remarks.

Seismic Design Analysis

___ Only SRE is in the room; therefore, there are no seismically induced failures (Category 2/1 interactions).
_X_ The NSRE in the room is seismically restrained (upgraded to Category 1 requirements); therefore, no seismically induced failures of these restraints are postulated.
___ The NSRE, which is postulated to fail as a result of an SSE, does not impact SRE.
___ Other, see remarks.

Missile Analysis

___ No credible missile sources exist in the room.
___ Missiles from rotating components generated within the room contain insufficient energy to escape their equipment housing(s).
_X_ Other, see remarks.

Pipe Break Analysis

_X_ There are no high-energy lines in the room.
___ The high-energy line breaks have been evaluated and do not adversely affect SRE.
___ Other, see remarks.

Remarks

1. Missiles from sources within the room, coincident with the most limiting single active failure (assumed in accordance with paragraphs 3.6.1.1.F and 3.6.1.1.G), will not preclude mitigation of the event or safe shutdown of the plant.

TABLE 3F-1 (SHEET 83 OF 84)

Listing of safety-related items in room R-D130

| Item Number | Description | Equipment Designation | Essential Equipment | Seismic Category 1 |
|---|---|---|---|---|
| 1. | Train B raceway | Conduits and cabletrays | Y | Y |
| 2. | Air-handling unit | 1-1555-A7-008 | Y | Y |

TABLE 3F-1 (SHEET 84 0F 84)

Rooms: R-C60; R-C106; R-C110; R-C125; R-C130

Effects analysis:

Pressure/temperature - There are no high-energy line breaks located in the above rooms. As the above rooms do not have high-energy line breaks and are not break nodes, they are included as a flowpath in the nodal model used in the pressure/temperature analysis of the steam generator blowdown system.

The above rooms are affected by pressure/temperature analysis.

Equipment is designed to withstand the effects of pressure, temperature, and humidity. For the pressure/temperature/humidity data, see table 3.11.B.1-1.

Structures are designed to withstand the short-term pressure/ temperature effects of high-energy line breaks.

TABLE 3F-2

UNIT 1 AFW PUMP HOUSE DESIGN PARAMETERS

| Node No.[a] | Initial Conditions | | | Design Conditions | | | | | | | | | | Blowdown | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Pressure (psia) | Temp. (°F) | Rel. Hum. (%) | Pressure (psia) | Temp.[b] (°F) | Rel. Hum. (%) | Flood Level (in.) | Nodal Volume ($ft^3$) | Nodes[a] | Press. Flow Area ($ft^2$) | Calc. after Break (psia) | Peak Temp. (°F) | Time (s) | Mass Flowrate (lb/s) | Enthalpy (Btu/lb) |
| | | | | | | | | | | | | | Steam Line | | |
| 1. | 14.7 | 120 | 50 | 17.7 | 320 | 0-100 | (c) | $12.6 \times 10^3$ | 1-2 | 24.0 | 15.4 | 184 | 0 | 0 | 0 |
| 2. | 14.7 | 120 | 50 | 17.7 | 320 | 0-100 | (d) | $10.9 \times 10^3$ | 1-3 | 20.0 | 15.5 | 249 | 0.004 | 237.4 | 1188.6 |
| | | | | | | | | | | | | | 0.01 | 210.5 | 1156.7 |
| 3. (atm) | 14.7 | 120 | 50 | - | - | - | - | - | - | - | - | - | 0.1 | 143.8 | 1187.3 |
| | | | | | | | | | | | | 0.2 | 126.7 | 1191.1 | |
| | | | | | | | | | | | | 0.5 | 110.8 | 1191.7 | |
| | | | | | | | | | | | | 0.9 | 104.5 | 1191.3 | |
| | | | | | | | | | | | | 2.0 | 100.9 | 1190.4 | |
| | | | | | | | | | | | | 20 | 100.7 | 1190.1 | |
| | | | | | | | | | | | | | Auxiliary Feedwater | | |
| | | | | | | | | | | | | | 0 | 0 | |
| | | | | | | | | | | | | | 60 | 282 | |
| | | | | | | | | | | | | | 1800 | 142 | |

a. Refer to drawing 1X6DD300 for nodal boundary.

b. The design of the structure assumes a 2-h duration for peak temperature.

c. Node 1 Flood level (in.):
Room 104 - 86
Room 105 - 97.2

d. Node 2 Flood level (in.):
Room 106 - 11

TABLE 3F-3A (SHEET 1 OF 4)

UNIT 1 MSIV/MFIV ROOM FLOODING (CASE 2) AND PRESSURE (CASE 1) ANALYSIS

| | Initial Conditions | | | Design Conditions | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Node[a] No. | Pressure (psia) | Temp. (°F) | Rel. Hum. (%) | Pressure (psia) | Temp (°F) | Rel. Hum. (%) | Flood Level (in.) | Node Volume ($ft^3$) | Calc Peak Pressure (psia) |
| I. Auxiliary Building MSIV/MFIV Area | | | | | | | | | |
| 1 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 5520.0 | 18.4 |
| 2 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 17423.0 | 18.0 |
| 3 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 7 | 3505.5 | 19.1 |
| 4 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 67 | 6033.88 | 19.4 |
| 5 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 4138.8 | 20.3 |
| 6 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 67 | 4237.0 | 19.4 |
| 7 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 7 | 1989.8 | 18.8 |
| 8 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 12684.8 | 18.0 |
| 9 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 67 | 22883.9 | 20.8 |
| 10 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 9577.0 | 24.4 |
| 11 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 138 | 5415.0 | 23.5 |
| 12 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 4651.2 | 18.8 |
| 13 | atm | | | | | | | | |

TABLE 3F-3A (SHEET 2 OF 4)

| Node[a] No. | Initial Conditions Pressure (psia) | Temp. (°F) | Rel. Hum. (%) | Design Conditions Pressure (psia) | Temp (°F) | Rel. Hum. (%) | Flood Level (in.) | Node Volume ($ft^3$) | Calc Peak Pressure (psia) |
|---|---|---|---|---|---|---|---|---|---|
| II. Auxiliary Building MSIV/MFIV Area Outside Restraint Wall | | | | | | | | | |
| 1 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 5520.0 | 17.5 |
| 2 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 17423.0 | 17.3 |
| 3 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 7 | 3505.5 | 17.4 |
| 4 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 67 | 6033.88 | 17.5 |
| 5 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 4138.8 | 17.2 |
| 6 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 67 | 4273.0 | 17.9 |
| 7 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 7 | 1989.8 | 17.9 |
| 8 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 12684.8 | 17.4 |
| 9 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 67 | 22883.9 | 17.4 |
| 10 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 9577.0 | 24.4 |
| 11 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 138 | 5415.0 | 23.5 |
| 12 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 4651.2 | 18.8 |
| 13 | atm | | | | | | | | |

TABLE 3F-3A (SHEET 3 OF 4)

| Node[a] No. | Initial Conditions Pressure (psia) | Temp. (°F) | Rel. Hum. (%) | Design Conditions Pressure (psia) | Temp (°F) | Rel. Hum. (%) | Flood Level (in.) | Node Volume ($ft^3$) | Calc Peak Pressure (psia) |
|---|---|---|---|---|---|---|---|---|---|
| III. Control Building MSIV/MFIV Area | | | | | | | | | |
| 1 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 3079.9 | 21.4 |
| 2 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 17617.0 | 20.2 |
| 3 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 22 | 11295.2 | 20.4 |
| 4 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 3769.6 | 19.8 |
| 5 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 21514.5 | 19.5 |
| 6 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 22 | 13740.5 | 20.3 |
| 7 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | NA | 8853.6 | 16.8 |
| 8 | atm | | | | | | | | |
| IV. Control Building MSIV/MFIV Area Outside Restraint Wall | | | | | | | | | |
| 1 | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 16 | 28805.2 | 23.5 |

a. Refer to figure 3F-1 (sheets 1 through 4) for nodal boundary.

TABLE 3F-3A (SHEET 4 OF 4)

Blowdown Data (Case 1)
(Break at Node 10)

| Time (s) | Mass Flowrate (lbm/s) | Enthalpy (BTU/lbm) |
|---|---|---|
| 0.0 | 0 | 1188 |
| 0.004 | 18,959 | 1138 |
| 0.008 | 19,905 | 1139 |
| 0.01 | 19,399 | 1137 |
| 0.03 | 17,567 | 1136 |
| 0.06 | 16,747 | 1139 |
| 0.10 | 16,030 | 1140 |
| 0.15 | 15,450 | 1140 |
| 0.20 | 15,312 | 1142 |
| 1.0 | 12,994 | 1153 |
| 1.2 | 12,378 | 1141 |
| 1.4 | 9304 | 1119 |
| 1.6 | 14,163 | 814 |
| 1.65 | 15,040 | 791 |
| 1.80 | 17,716 | 725 |
| 1.90 | 18,414 | 712 |
| 2.00 | 18,831 | 706 |
| 2.15 | 19,854 | 700 |
| 2.30 | 19,597 | 695 |
| 3.0 | 17,259 | 688 |
| 3.4 | 21,428 | 668 |
| 4.0 | 20,755 | 661 |
| 4.8 | 20,415 | 653 |

TABLE 3F-3B (SHEET 1 OF 2)

UNIT 1 CONTROL BUILDING MSIV/MFIV VAULT AREA MSLB TEMPERATURE (CASE 3) ANALYSIS

| | | Initial Conditions | | | Design Conditions | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Node No.[a] | Description | Pressure (psia) | Temp. (°F) | Rel. Hum. (%) | Pressure (psia) | Temp. (°F) | Rel. Hum. (%) | Volume ($ft^3$) | Node Elev. (ft) | Calc. Peak[c] Temp. (°F) |
| 1 | West Steam Line Room (Lower) | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 18,361.4 | 224.0 | 483.2[c] |
| 2 | West Aux. Feedline Room | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 11,300.0 | 200.5 | 399.0 |
| 3 | East SL Room (Lower) | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 11,370.0 | 224.0 | 413.1 |
| 4 | East AFL Room | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 13,740.0 | 200.5 | 360.9 |
| 5 | Entrance Room (Lower) | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 4,234.0 | 221.0 | 390.2 |
| 6 | Entrance Room (Upper) | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 4,234.0 | 239.0 | 398.7 |
| 7 | East Penthouse | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 3,208.0 | 257.0 | 428.3 |
| 8 | West Penthouse | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 2,338.6 | 257.0 | 477.4 |
| 9 | East SL Room (Upper) | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 11,370.0 | 240.5 | 429.1 |
| 10 | West Steam Line Room (Upper) | 14.7 | 120 | 50 | 29.7 | 320 | 0-100 | 9,180.0 | 240.5 | 478.3 |

TABLE 3F-3B (SHEET 2 OF 2)

| Flow Path Data | | | | Blowdown Data (1.0 $ft^2$ MSLB[b] at 102% Power) | | |
|---|---|---|---|---|---|---|
| Flow Path No. | Upstream Node | Downstream Node | Flow Area $ft^2$ | Time (s) | Mass Flow-rate (lbm/s) | Enthalpy (Btu/lbm) |
| 1 | 4 | 3 | 145.0 | 0.0 | 0.0 | 0.0 |
| 2 | 2 | 1 | 139.0 | 1.5 | 1980.0 | 1192.0 |
| 3 | 1 | 3 | 218.6 | 6.5 | 1904.0 | 1194.0 |
| 4 | 2 | 4 | 245.0 | 14.0 | 2282.0 | 1188.0 |
| 5 | 3 | 5 | 95.0 | 15.0 | 2283.0 | 1189.0 |
| 6 | 9 | 6 | 144.0 | 15.5 | 2173.0 | 1191.0 |
| 7 | 9 | 7 | 139.5 | 23.5 | 1490.0 | 1201.0 |
| 8 | 1 | 8 | 120.0 | 33.0 | 1187.0 | 1204.0 |
| 9 | 1 | 9 | 205.5 | 55.5 | 924.2 | 1204.0 |
| 10 | 3 | 9 | 876.6 | 64.5 | 762.0 | 1236.0 |
| 11 | 1 | atm | 4.5 | 81.5 | 203.9 | 1288.0 |
| 12 | 2 | atm | 6.85 | 90.5 | 100.3 | 1298.0 |
| 13 | 3 | atm | 4.5 | 123.5 | 78.0 | 1304.0 |
| 14 | 4 | atm | 6.85 | 162.5 | 78.0 | 1305.0 |
| 15 | 5 | atm | 141.0 | 1800.0 | 78.0 | 1287.0 |
| 16 | 6 | atm | 141.0 | 1812.0 | 12.5 | 1295.0 |
| 17 | 7 | atm | 164.35 | | | |
| 18 | 8 | atm | 135.85 | | | |
| 19 | 5 | 6 | 268.0 | | | |
| 20 | 1 | break | 1.0 | | | |
| 21 | 1 | 10 | 725.0 | | | |

a. Refer to figure 3F-2 for nodal diagrams.

b. Main steam line break.

c. The calculated peak temperatures correspond to a 1.0 $ft^2$ MSLB at 102% power (102% of 3579 MWt). This break resulted in the overall highest peak temperature which occurred in break node 1. The peak temperatures and pressures for other nodes actually may be higher for other breaks.

TABLE 3F-4 (SHEET 1 OF 3)

EVALUATION OF RCS LOOP BRANCH LINE BREAKS

| RCS Loop Nozzle No. | Branch Line Description | Branch Line Identification | Pipe Break Evaluation |
|---|---|---|---|
| Loop 1 cold leg | | | |
| Nozzle - 5 | Boron injection tank (safety injection system (SIS)) | 1204-243-1£ in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 14 | Accumulator injection | 1204-124-10 in. | Pipe whip: Two restraints were installed to prevent pipe whip.(b) Jet impingement: One barrier was installed to prevent the failure of any essential components.(b) |
| | Upstream of valve 083 | 1204-120-10 in.<br>1204-042-6 in.<br>1204-036-2 in. | Pipe whip: Two restraints were installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 11 | CVCS normal charging line | 1208-009-3 in. to valve 035 (second check valve) | Pipe whip: Three restraints were installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| | CVCS normal charging upstream of valve 035(second valve) | 1208-008-3 in. | Pipe whip: One restraint was installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 12 | Pressurizer spray header | 1201-029-4 in. | Pipe whip: One restraint was installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 1 crossover leg | | | |
| Nozzle - 18 | Reactor coolant loop drain | 1201-031-2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 1 hot leg | | | |
| Nozzle - 15 | RHR system recirculation/SIS | 1201-036-12 in. (b) | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |

TABLE 3F-4 (SHEET 2 OF 3)

| RCS Loop Nozzle No. | Branch Line Description | Branch Line Identification | Pipe Break Evaluation |
|---|---|---|---|
| Loop 2 cold leg | | | |
| Nozzle - 5 | Boron injection tank (SIS) | 1204-244-1 1/2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 14 | Accumulator injection | 1204-125-10 in.[(b)] | Pipe whip: Two restraints were installed to prevent pipe whip.[(b)] Jet impingement: Breaks do not result in the failure of any essential components.[(b)] |
| | Upstream of valve 084 | 1204-121-10 in.<br>1204-043-6 in.<br>1204-035-2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 2 crossover leg | | | |
| Nozzle - 18 | Reactor coolant loop drain | 1201-042-2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 2 hot leg | | | |
| Nozzle - 13 | RHR SIS injection | 1204-024-6 in. | Pipe whip: One restraint was installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 3 cold leg | | | |
| Nozzle - 5 | Boron injection tank (SIS) | 1204-245-1 1/2 in. | Pipe whip: Pipe whips but does result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 14 | Accumulator injection | 1204-126-10 in.[(b)] | Pipe whip: Two restraints were installed to prevent pipe whip.[(b)] Jet impingement: Breaks do not result in the failure of any essential components.[(b)] |
| | Upstream of valve 085 | 1204-122-10 in.<br>1204-044-6 in.<br>1204-034-2 in. | Pipe whip: Two restraints were installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 4 | Letdown (CVCS) | 1201-048-3 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 3 crossover leg | | | |
| Nozzle - 18 | Reactor coolant loop drain | 1201-046-2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |

TABLE 3F-4 (SHEET 3 OF 3)

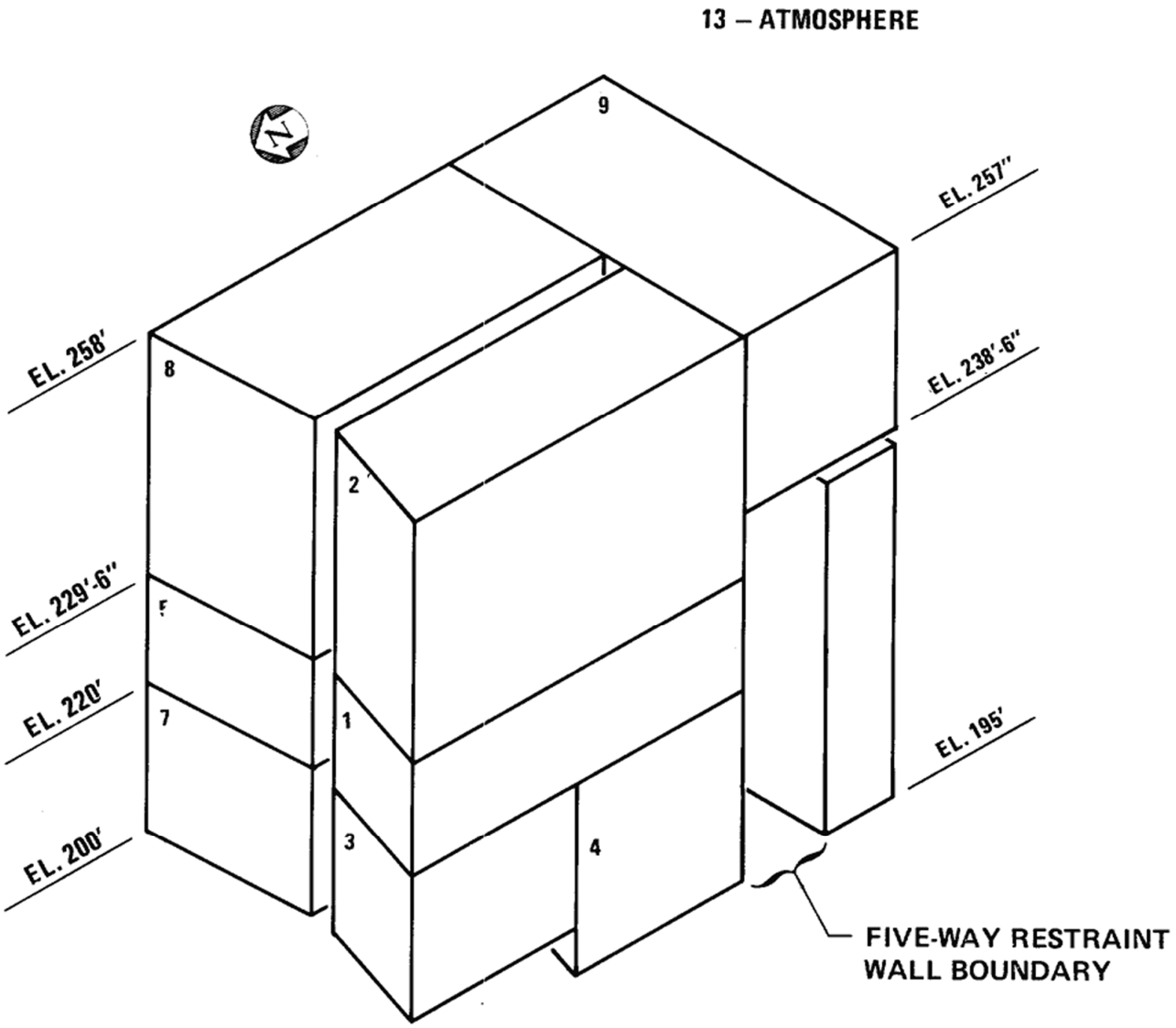
| RCS Loop Nozzle No. | Branch Line Description | Branch Line Identification | Pipe Break Evaluation |
|---|---|---|---|
| Loop 3 hot leg | | | |
| Nozzle - 13 | RHR SIS injection | 1204-025-6 in. | Pipe whip: One restraint was installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 4 cold leg | | | |
| Nozzle - 5 | Boron injection tank (SIS) | 1204-246-1 1/2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 14 | Accumulator injection | 1204-127-10 in.[(b)] | Pipe whip: Two restraints were installed to prevent pipe whip.[(b)] Jet impingement: One barrier was installed to prevent the failure of any essential components.[(b)] |
| | Upstream of valve 086 | 1204-123-10 in.[(b)]<br>1204-045-6 in.<br>1204-033-2 in. | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 12 | Pressurizer spray header | 1201-030-4 in.<br>1201-030-6 in.<br>1208-012-2 in. | Pipe whip: One restraint was installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Nozzle - 11 | CVCS alternate charging line | 1208-007-3 in. to valve 037 (second check valve) | Pipe whip: Three restraints were installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| | Upstream of valve 037 (second valve) | 1208-488-3 in. | Pipe whip: One restraint was installed to prevent pipe whip. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 4 crossover leg | | | |
| Nozzle - 18 | Reactor coolant loop drain | 1201-051-2 in. with excess letdown | Pipe whip: Pipe whips but does not result in the failure of any essential components. Jet impingement: Breaks do not result in the failure of any essential components. |
| Loop 4 hot leg | | | |
| Nozzle - 15 | RHR system recirculation/SIS injection | 1201-049-12 in.[(b)]<br>1204-021-6 in. | Pipe whip: One restraint was installed. Pipe whips but does not result in failure of any essential components. Jet impingement: One barrier was installed to prevent the failure of any essential components.[(b)] |

a. See drawing AX6DD309.
b. Applies to Unit 1 only.

TABLE 3F-5

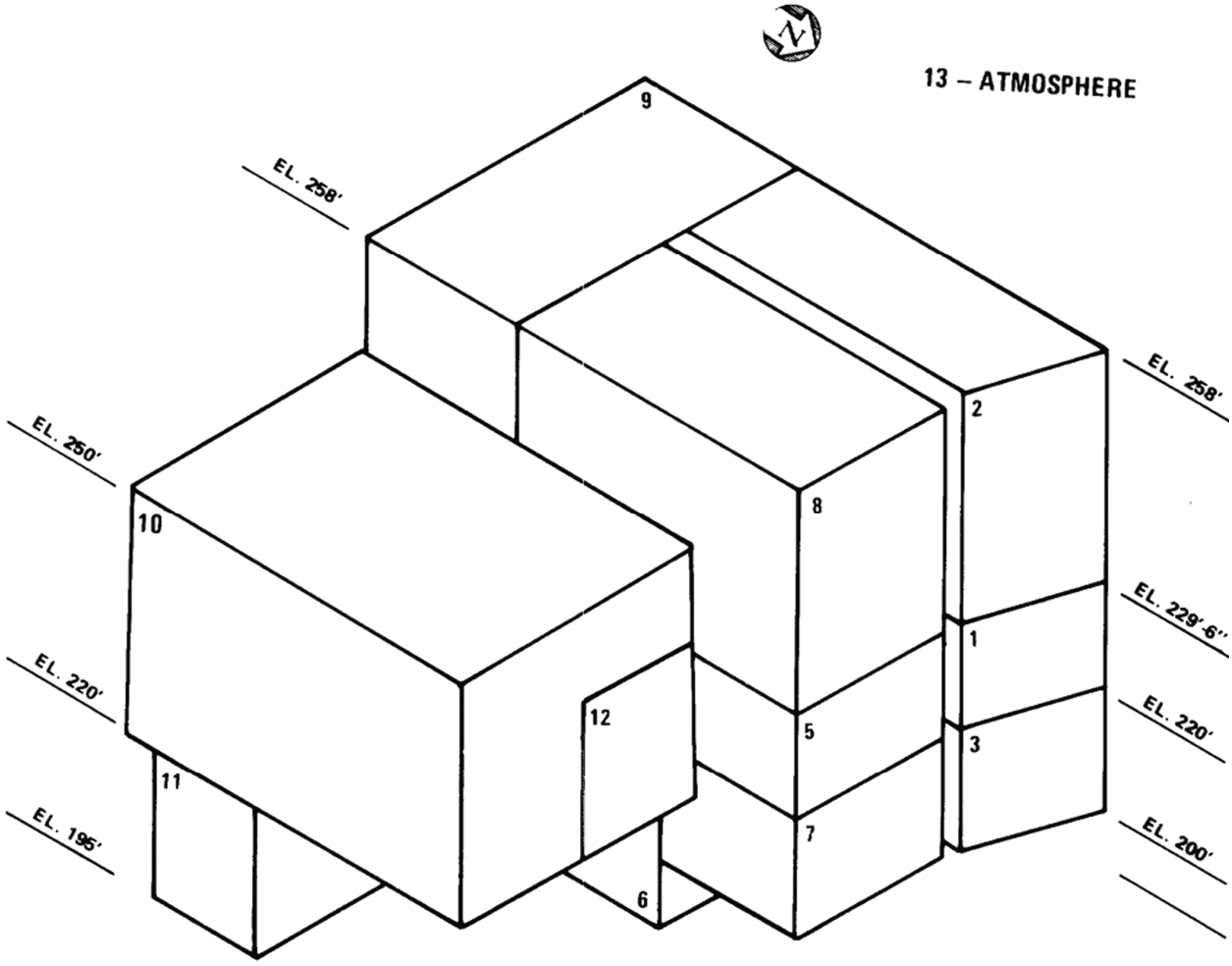
JET IMPINGEMENT BARRIERS

| System Line Number | Break Location | Target Number | Barrier Number | Building |
|---|---|---|---|---|
| Accumulator injection loop-4 1204-127-10 in. | P-14DB-0411-L-1 P-14DB-0411-L-2 | 1201-029-4 in. | BC-3 | Containment (Unit 1 only) |
| Main safety loops 1 and 4 from auxiliary building to main steam tunnel 1301-008-38 in. | P-17CA-1059-C | 1305-056-26 in. | BT-2 | Tunnel IT1 |
| Main feedwater loops 1-4 (tunnel to turbine building) 1305-055-36 in. | P-12EA-1135-C | 1301-007-36 in. | BT-1 | Tunnel IT1 |



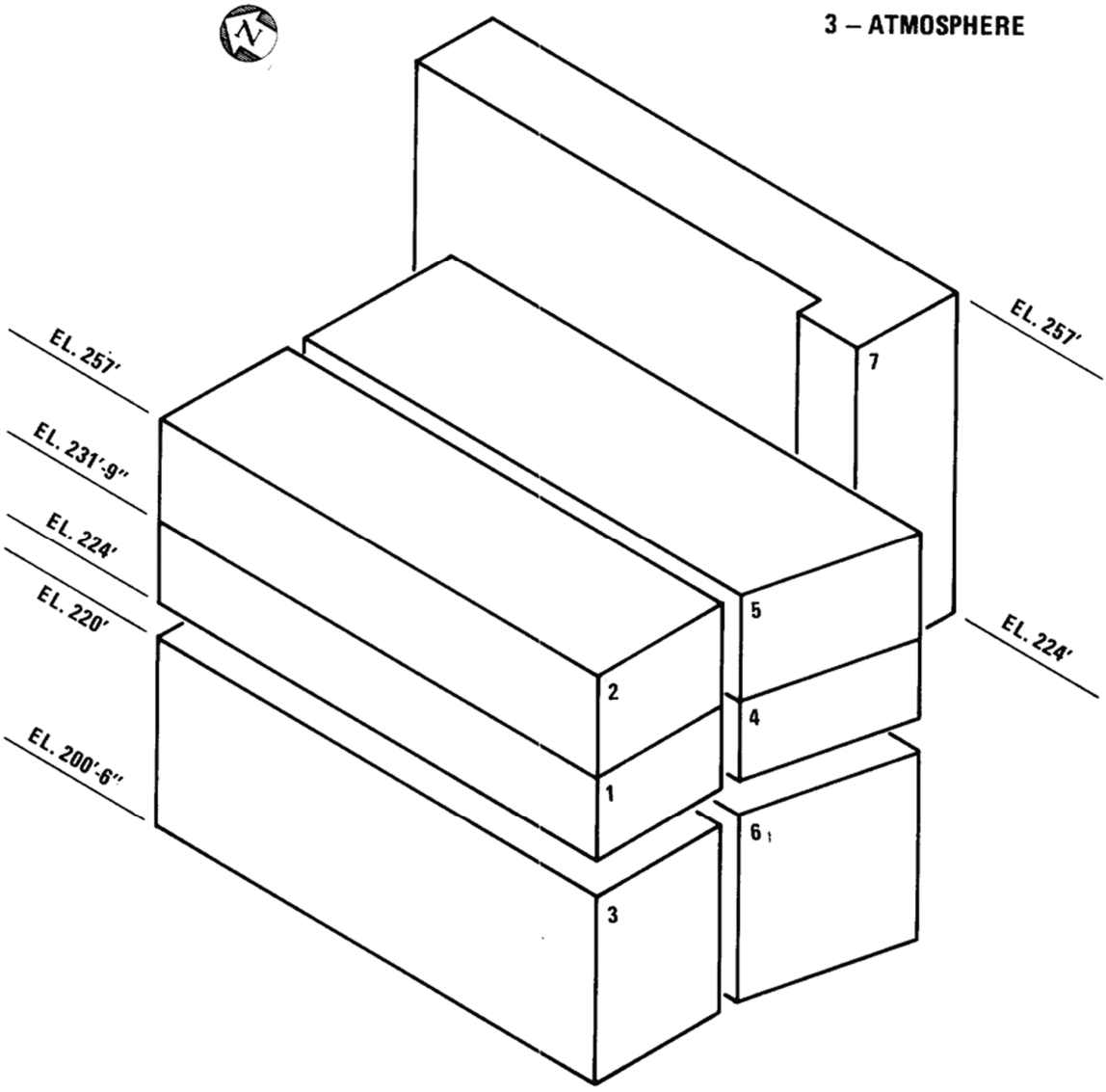

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| | FIGURE 3F–1 (SHEET 1 OF 4) |



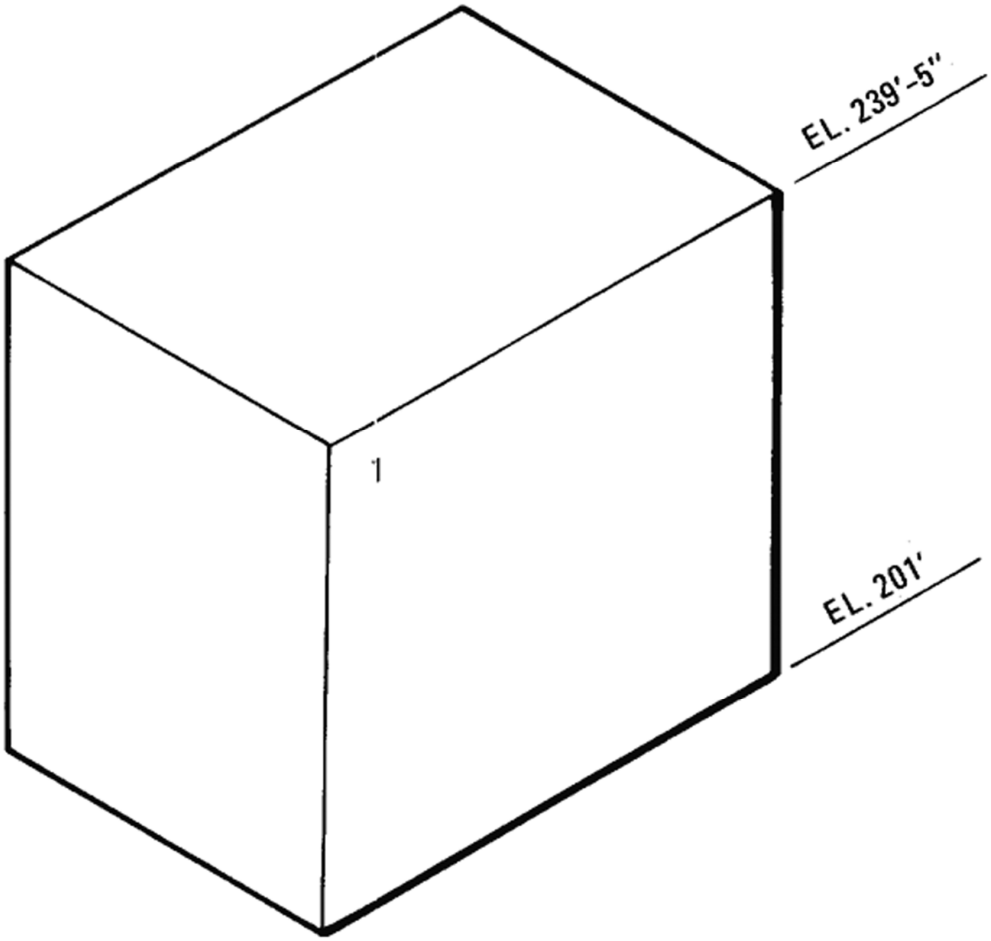

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| SOUTHERN COMPANY *Energy to Serve Your World®*<br>**VOGTLE**<br>**ELECTRIC GENERATING PLANT**<br>**UNIT 1 AND UNIT 2** | NODAL BOUNDARY FOR AUXILIARY BUILDING – MSIV/MFIV ROOMS |
| | FIGURE 3F–1 (SHEET 2 OF 4) |



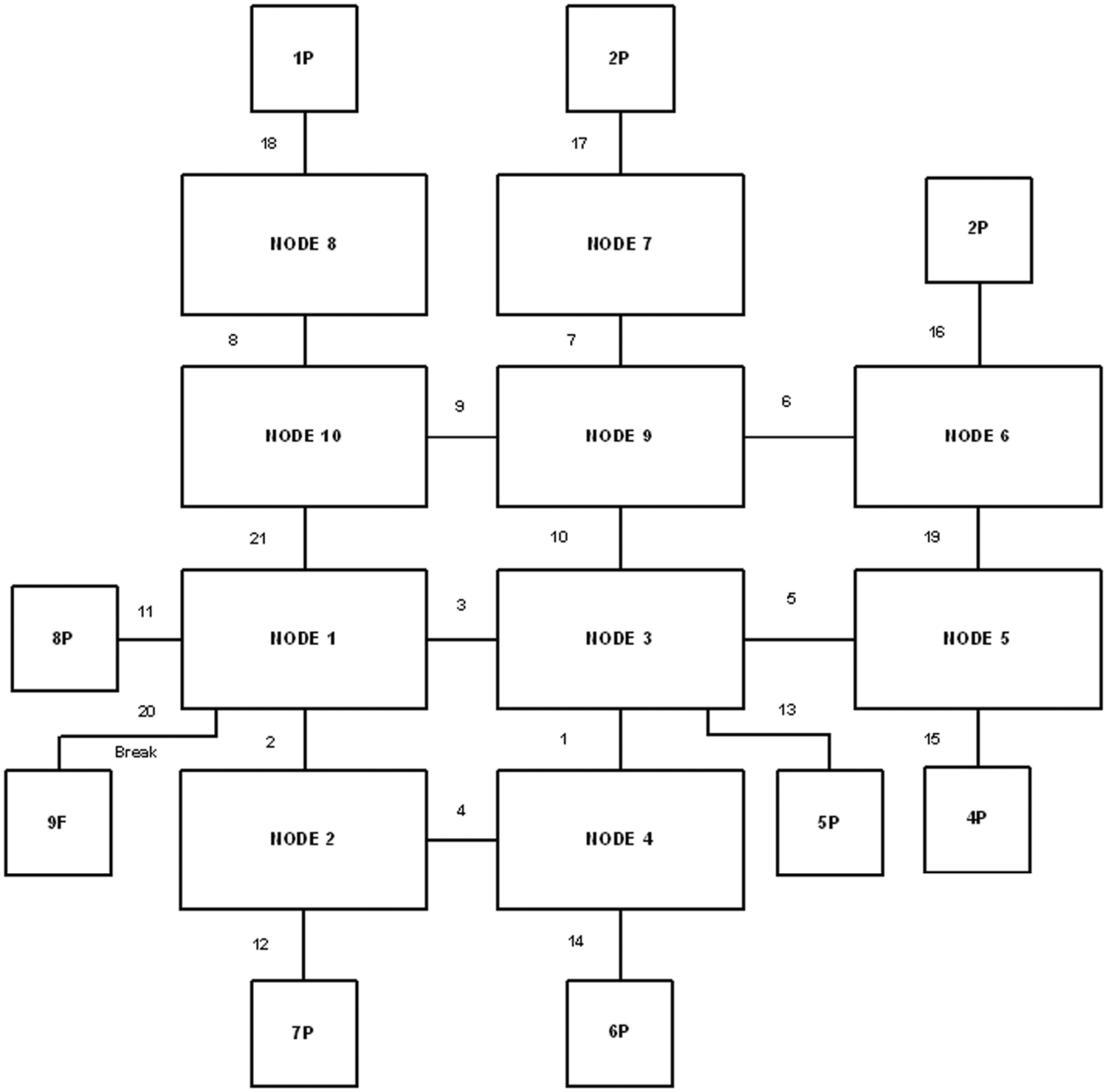

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|---|---|
| SOUTHERN COMPANY *Energy to Serve Your World®*<br>**VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2** | NODAL BOUNDARY FOR CONTROL BUILDING – MSIV/MFIV ROOMS (INSIDE RESTRAINT WALL) |
| | FIGURE 3F–1 (SHEET 3 OF 4) |



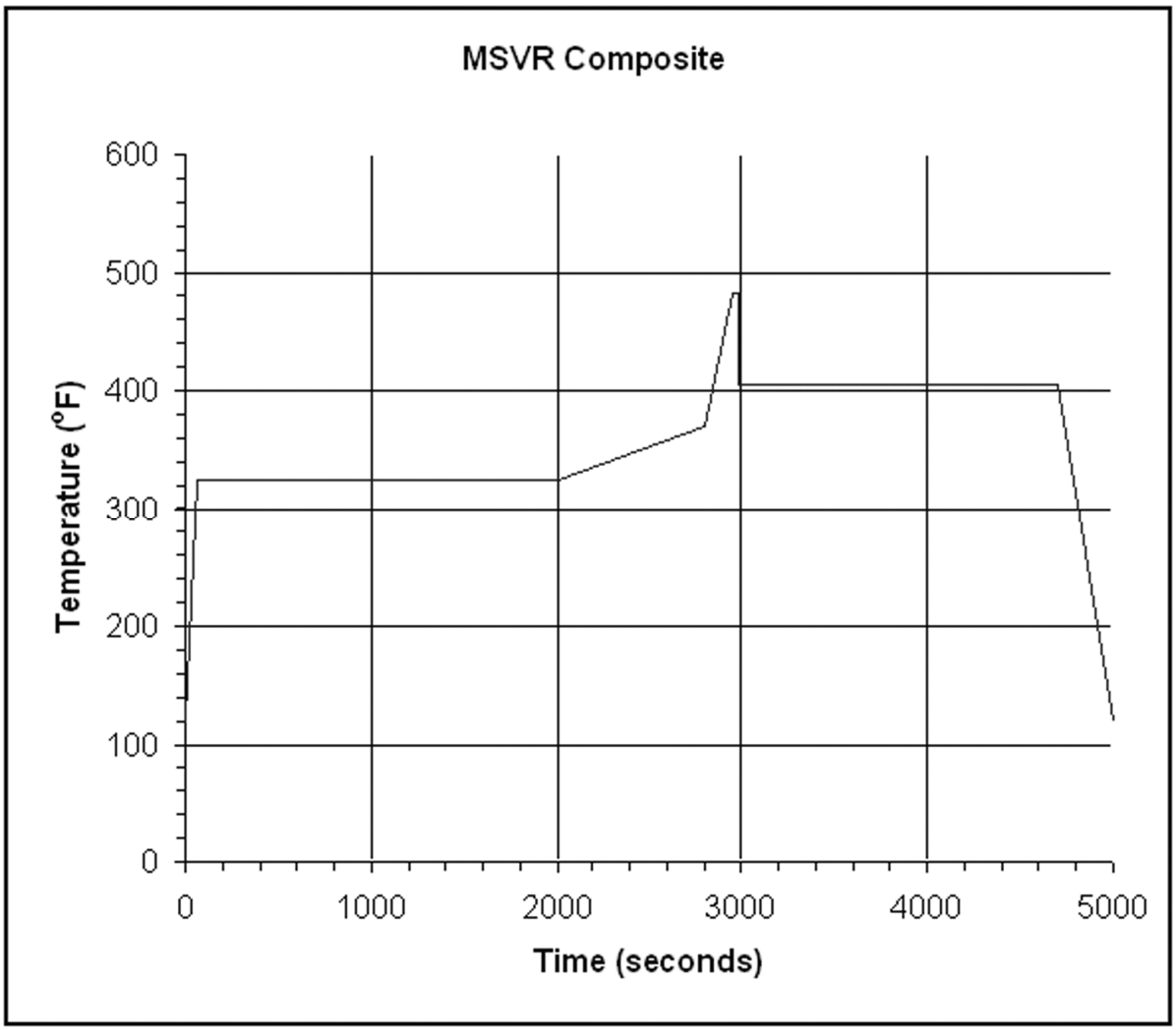

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| SOUTHERN COMPANY *Energy to Serve Your World®*<br>**VOGTLE**<br>**ELECTRIC GENERATING PLANT**<br>**UNIT 1 AND UNIT 2** | NODAL BOUNDARY FOR CONTROL BUILDING – MSIV/MFIV ROOMS (OUTSIDE RESTRAINT WALL) |
| | FIGURE 3F–1 (SHEET 4 OF 4) |




**REV 14 10/07**

| | |
|---|---|
| SOUTHERN COMPANY *Energy to Serve Your World®*<br>**VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2** | NODAL DIAGRAM FOR MSIV/MFIV GOTHIC ANALYSIS<br>FIGURE 3F–2 |




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SOUTHERN COMPANY
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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

TIME–HISTORY PLOT MAXIMUM
COMPARTMENT COMPOSITE TEMPERATURE
FOR MSIV AREA BREAK NODE

FIGURE 3F–3